

```
clear
% ECSE 563 assignmenet 3
% Ali Seifeldin
% https://github.com/Bakalala/MGCILL-ECSE-563
```

```
% Question 1.a
```

```
%  $R = \Delta f / \Delta P_e$ 
```

```
% 5% decrease in f, 100% increase in active P
```

```
f = 60 %Hz
```

```
f =  
60
```

```
Pe1 = 500 % MW
```

```
Pe1 =  
500
```

```
Pe2 = 250 % MW
```

```
Pe2 =  
250
```

```
PeSystem = 2000 % MW
```

```
PeSystem =  
2000
```

```
R1 = f * .05 / Pe1 % Hz/Mw
```

```
R1 =  
0.0060
```

```
R2 = f * .05 / Pe2 % Hz/Mw
```

```
R2 =  
0.0120
```

```
R1Hzpu = R1 * PeSystem % Hz/p.u.
```

```
R1Hzpu =  
12
```

```
R2Hzpu = R2 * PeSystem % Hz/p.u.
```

```
R2Hzpu =  
24
```

```
% To get p.u / p.u, 2000 MW p.u., 60 Hz p.u.
```

```
R1pu = R1 * PeSystem / f % p.u./p.u.
```

```
R1pu =  
0.2000
```

```
R2pu = R2 * PeSystem / f % p.u./p.u.
```

```
R2pu =  
0.4000
```

```
% Question 1.b
```

```
% J = (2 Wk0) / w0^2 J/  
% M = w0 J / = (2 Wk0) / w0
```

```
w0 = 2 * pi * f % rad / s
```

```
w0 =  
376.9911
```

```
Wk0 = 21 * 10^9 % GJ
```

```
Wk0 =  
2.1000e+10
```

```
Wk0_MWs = Wk0 / 1e6; % MW.s
```

```
M = 2 * Wk0_MWs /w0 % MW s^2 /rad.
```

```
M =  
111.4085
```

```
Mpu = M / PeSystem % p.u. s^2 / rad
```

```
Mpu =  
0.0557
```

```
% Question 1.c
```

```
Tch1 = 3 %s
```

```
Tch1 =  
3
```

```
Tch2 = 8 %s
```

```
Tch2 =  
8
```

```
Tg1 = 0.25
```

```
Tg1 =  
0.2500
```

```
Tg2 = 0.50
```

```
Tg2 =  
0.5000
```

```
R1 = 0.0060 * 2 * pi % rad / s MW
```

```
R1 =  
0.0377
```

```
R2 = 0.0120 * 2 * pi % rad / s MW
```

```
R2 =  
0.0754
```

```
M = 1.1141e+08 / 10^6 % convert to MW s^2 / rad
```

```
M =  
111.4100
```

```
D = 20 / ( 2 * pi) % MW s / rad
```

```
D =  
3.1831
```

```
% we do the math in MW rads/s
```

```
mdl = 'ass3q1c';  
load_system(mdl);  
set_param(mdl, 'StopTime', '30');
```

```
%% Run simulation  
out = sim(mdl);
```

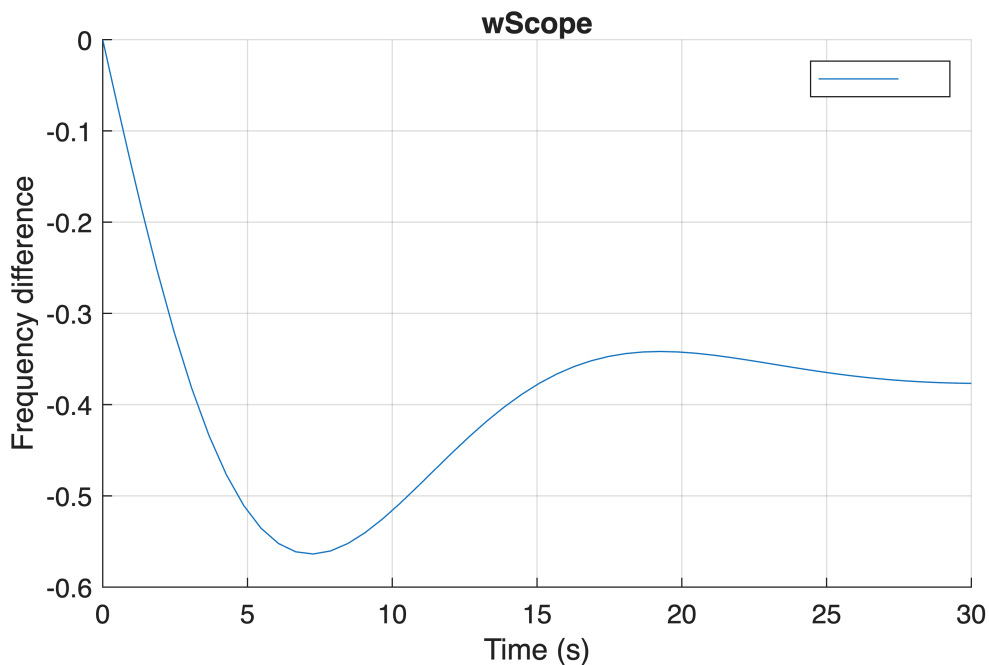
```
% Question 1.c
```

```
scopeNames = {'wScope'};  
  
for i = 1:numel(scopeNames)  
    name = scopeNames{i};  
  
    % Try to get dataset directly from 'out'  
    try  
        ds = out.get(name);  
    catch  
        % Otherwise get it from logouts  
        ds = out.logouts.get(name);  
    end
```

```

% Plot all signals in this dataset
figure('Name', name, 'NumberTitle', 'off');
hold on; grid on;
for j = 1:ds.numElements
    sig = ds.get(j);
    plot(sig.Values.Time, sig.Values.Data, 'DisplayName', sig.Name);
end
xlabel('Time (s)');
ylabel('Frequency difference');
title(name);
legend('show');
end

```



```

% we can see that the frequency drops and reaches a nadir of  $-0.55\text{Hz}$  and
% stabilises around  $-0.37\text{ Hz}$ 
% -->  $60 - 0.37\text{Hz}$  so  $59.63\text{Hz}$  is the steady state freq
% Simulation is run using rad / s MW

```

```

% Question 1.d

scopeNames = {'Pm1Scope', 'Pm2Scope', 'PScope'};

figure; hold on; grid on;

for i = 1:numel(scopeNames)
    name = scopeNames{i};

```



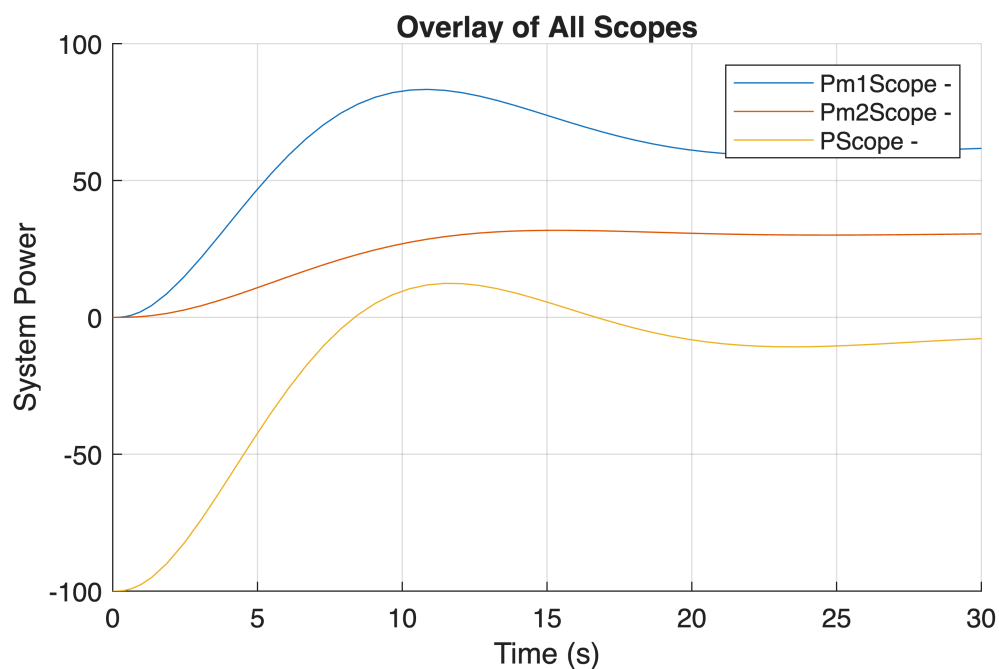
```

% Try to get dataset directly from 'out'
try
    ds = out.get(name);
catch
    % Otherwise get it from logout
    ds = out.logout.get(name);
end

% Plot each signal in this scope
for j = 1:ds.numElements
    sig = ds.get(j);
    plot(sig.Values.Time, sig.Values.Data, 'DisplayName', [name ' - '
sig.Name]);
end
end

xlabel('Time (s)');
ylabel('System Power');
title('Overlay of All Scopes');
legend('show', 'Interpreter', 'none');

```



```

% Pm1 and Pm2 start to ramp up, Pm1 ramps up much faster and does more
'effort' and then eventually
% stabilize to a 2/3 and 1/3 split. between both
% we can see on Pscope that with an initial -100 MW, it quickly shoots up
% to ~+20 MW and stabilizes close to the -10 MW.
% the results are consistent with the droop control

```

```

% Question 1.e

```

```
Tch1 = 3 %s
```

```
Tch1 =  
3
```

```
Tch2 = 8 %s
```

```
Tch2 =  
8
```

```
Tg1 = 0.25
```

```
Tg1 =  
0.2500
```

```
Tg2 = 0.50
```

```
Tg2 =  
0.5000
```

```
R1 = 0.0060 * 2 * pi % rad / s MW
```

```
R1 =  
0.0377
```

```
R2 = 0.0120 * 2 * pi % rad / s MW
```

```
R2 =  
0.0754
```

```
M = 1.1141e+08 / 10^6 % convert to MW s^2 / rad
```

```
M =  
111.4100
```

```
D = 20 / ( 2 * pi) % MW s / rad
```

```
D =  
3.1831
```

```
% we do the math in MW rads/s
```

```
mdl = 'ass3q1e';  
load_system(mdl);  
set_param(mdl, 'StopTime', '100');
```

```
%% Run simulation  
out = sim(mdl);
```

```
scopeNames = {'wScope'};
```

```
for i = 1:numel(scopeNames)
```

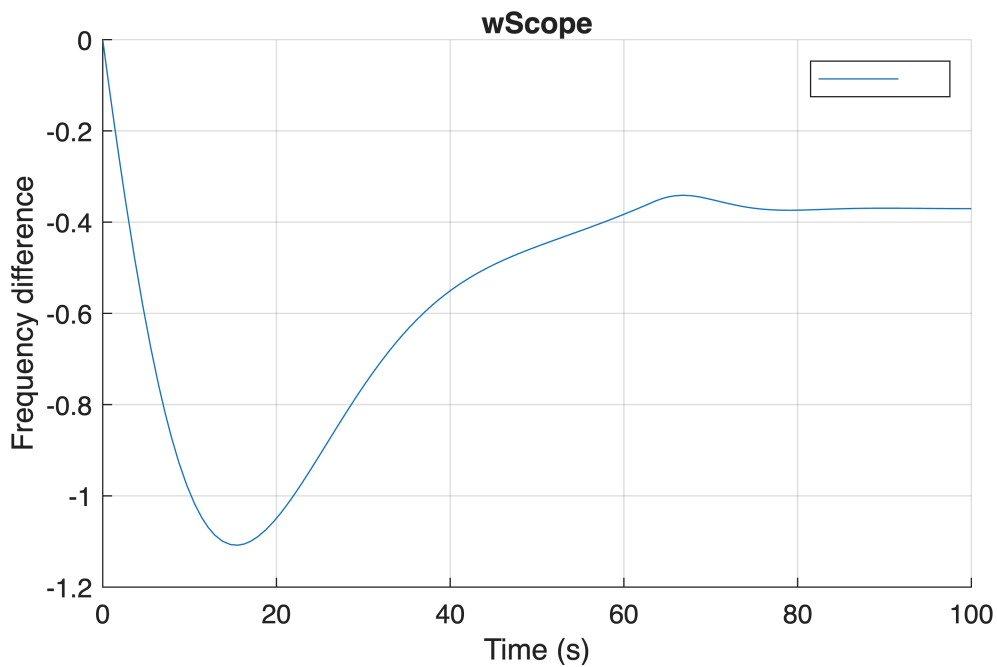
```

name = scopeNames{i};

% Try to get dataset directly from 'out'
try
    ds = out.get(name);
catch
    % Otherwise get it from logouts
    ds = out.logouts.get(name);
end

% Plot all signals in this dataset
figure('Name', name, 'NumberTitle', 'off');
hold on; grid on;
for j = 1:ds.numElements
    sig = ds.get(j);
    plot(sig.Values.Time, sig.Values.Data, 'DisplayName', sig.Name);
end
xlabel('Time (s)');
ylabel('Frequency difference');
title(name);
legend('show');
end

```



```

scopeNames = {'Pm1Scope', 'Pm2Scope', 'PScope'};

figure; hold on; grid on;

for i = 1:numel(scopeNames)
    name = scopeNames{i};

```

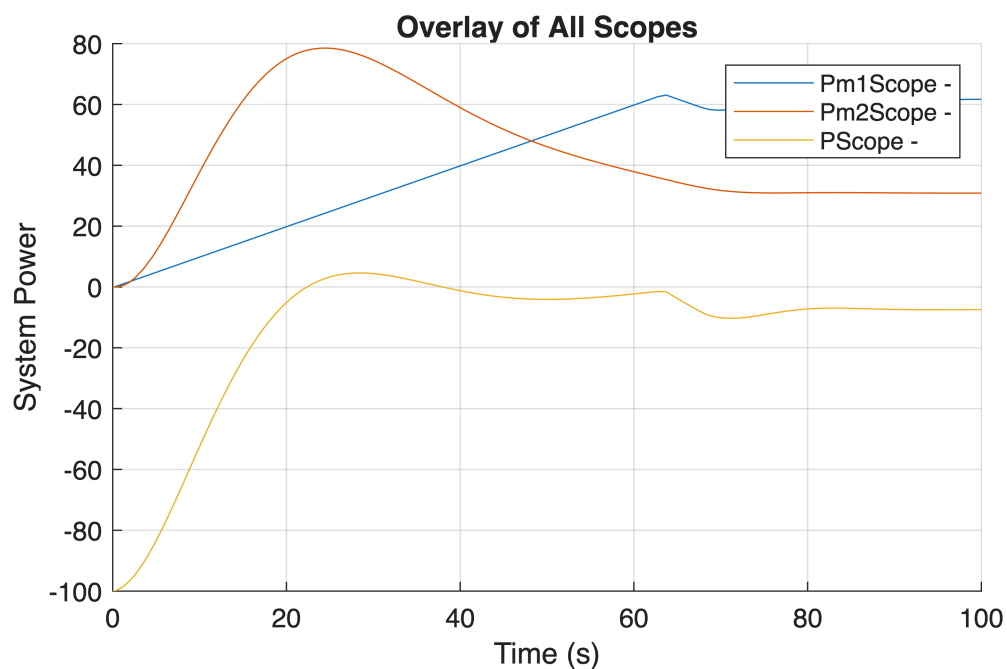
```

% Try to get dataset directly from 'out'
try
    ds = out.get(name);
catch
    % Otherwise get it from logout
    ds = out.logout.get(name);
end

% Plot each signal in this scope
for j = 1:ds.numElements
    sig = ds.get(j);
    plot(sig.Values.Time, sig.Values.Data, 'DisplayName', [name ' - '
sig.Name]);
end
end

xlabel('Time (s)');
ylabel('System Power');
title('Overlay of All Scopes');
legend('show', 'Interpreter', 'none');

```



```

% we can observe that setting a slew rate limit of 1 causes
% the frequency to have a nadir of ~-1.1 HZ so 58.9Hz. We also see that
% it takes longer for the freq stabilize.
% P2 does most of the work to compensate for the difference initially and
% then it goes down as P1 ramps up. the total power P stabilizes closer to
% 0 until 60s but then drops

```

```

% Question 2

```

```
% tuning K
```

```
Tch1 = 3 %s
```

```
Tch1 =  
3
```

```
Tch2 = 8 %s
```

```
Tch2 =  
8
```

```
Tg1 = 0.25
```

```
Tg1 =  
0.2500
```

```
Tg2 = 0.50
```

```
Tg2 =  
0.5000
```

```
R1 = 0.0060 * 2 * pi % rad / s MW
```

```
R1 =  
0.0377
```

```
R2 = 0.0120 * 2 * pi % rad / s MW
```

```
R2 =  
0.0754
```

```
M = 1.1141e+08 / 10^6 % convert to MW s^2 / rad
```

```
M =  
111.4100
```

```
D = 20 / ( 2 * pi) % MW s / rad
```

```
D =  
3.1831
```

```
K = 0.3
```

```
K =  
0.3000
```

```
% we do the math in MW rads/s
```

```
for i = 1:10
```

```
    mdl = 'ass3q2';  
    load_system(mdl);
```

```

set_param mdl, 'StopTime', '1000');

%% Run simulation
out = sim mdl;

scopeNames = {'wScope'};

for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logsout
        ds = out.logsout.get(name);
    end

    % Plot all signals in this dataset
    figure('Name', name, 'NumberTitle', 'off');
    hold on; grid on;
    for j = 1:ds.numElements
        sig = ds.get(j);
        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', sig.Name);
    end
    xlabel('Time (s)');
    ylabel('Frequency difference');
    title(name);
    legend('show');
end

scopeNames = {'Pm1Scope', 'Pm2Scope', 'PScope'};

figure; hold on; grid on;

for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logsout
        ds = out.logsout.get(name);
    end

    % Plot each signal in this scope
    for j = 1:ds.numElements
        sig = ds.get(j);

```

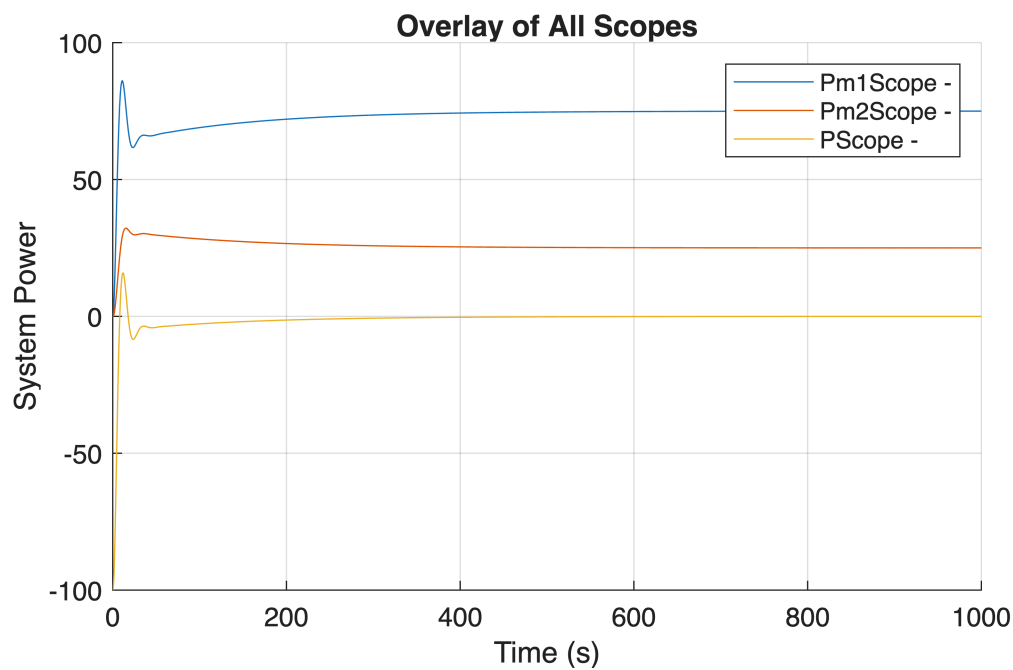
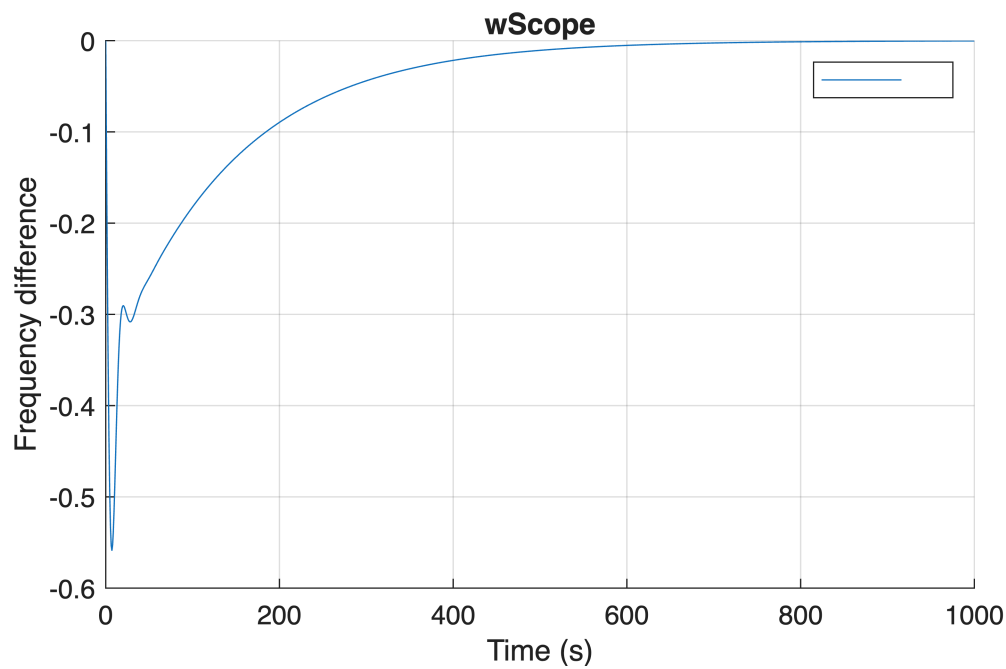
```

        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', [name ' -
' sig.Name]);
    end
end

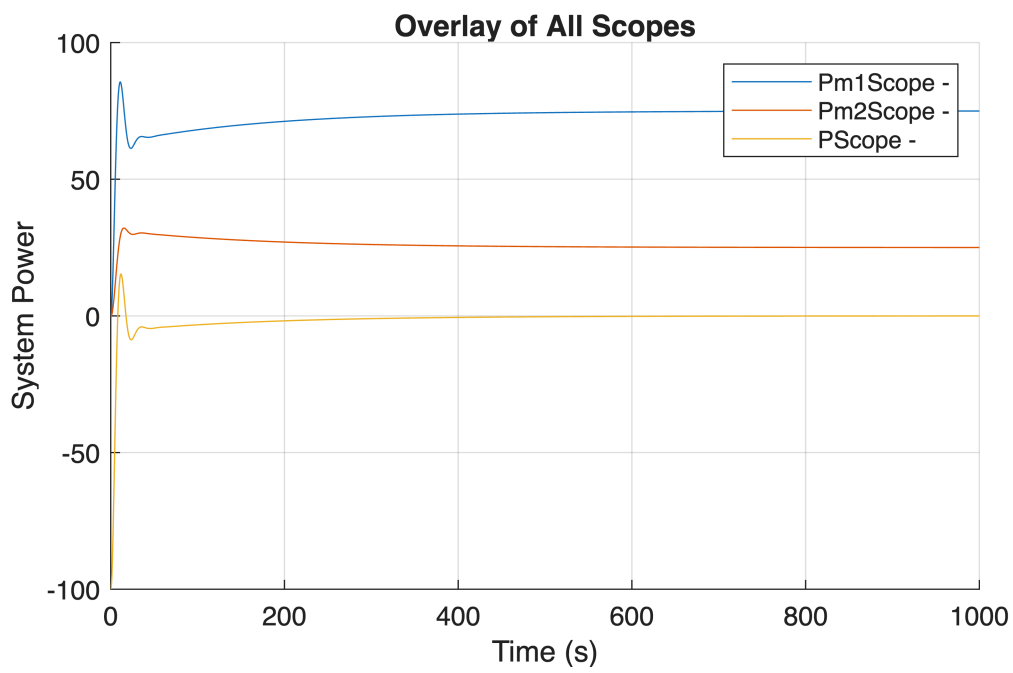
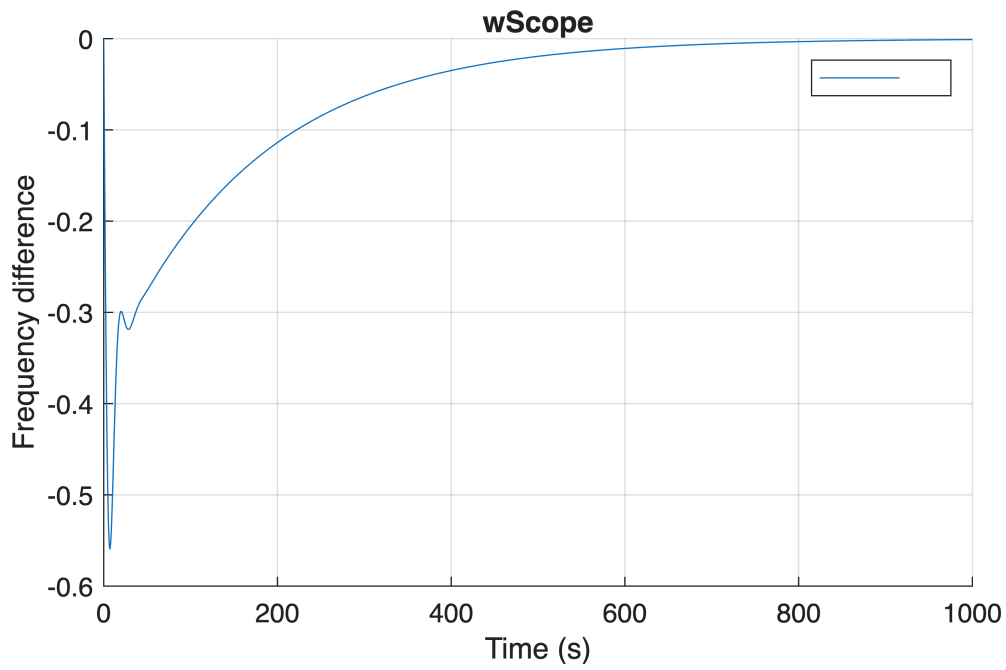
xlabel('Time (s)');
ylabel('System Power');
title('Overlay of All Scopes');
legend('show', 'Interpreter', 'none');

K = K - .05
end

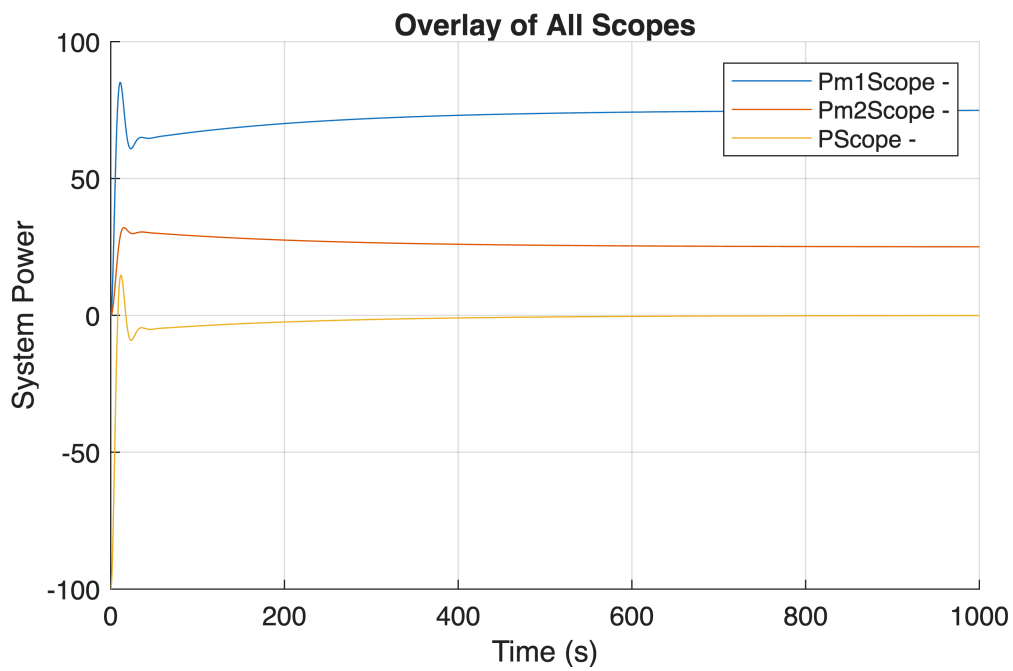
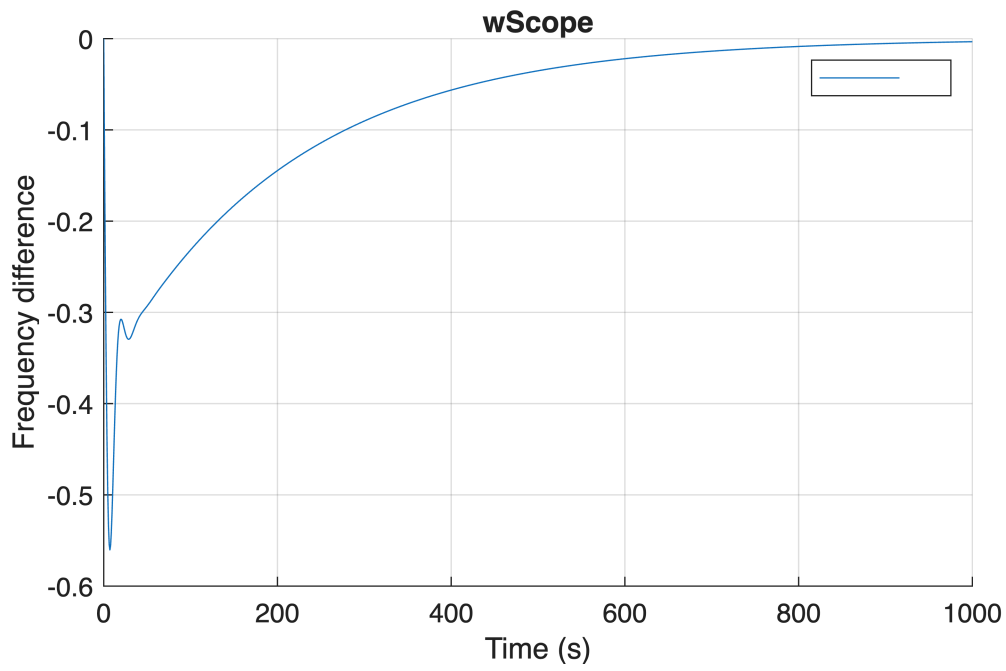
```



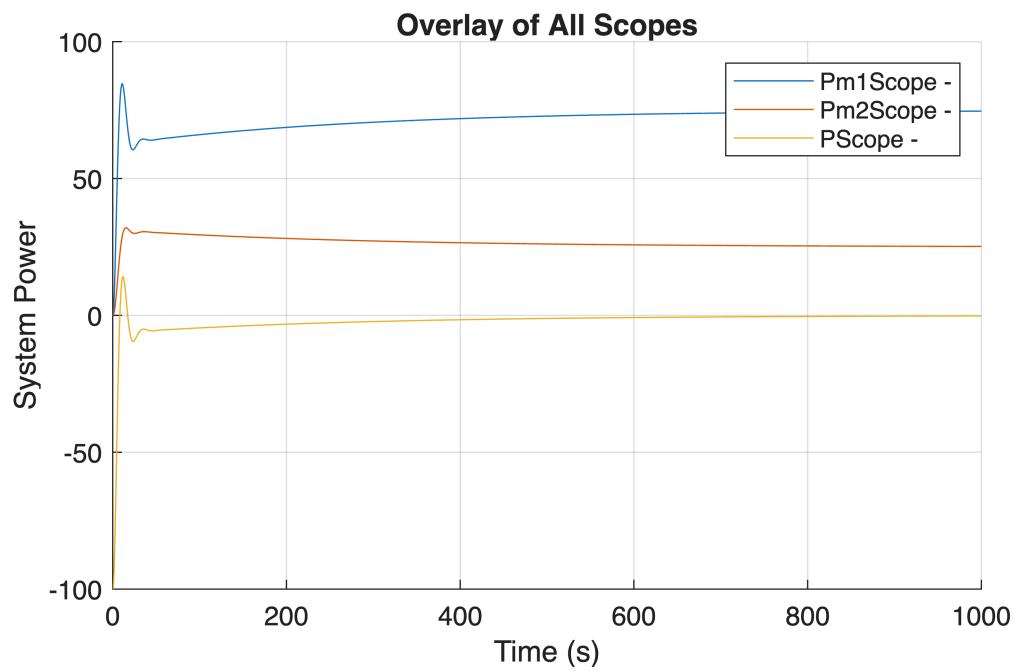
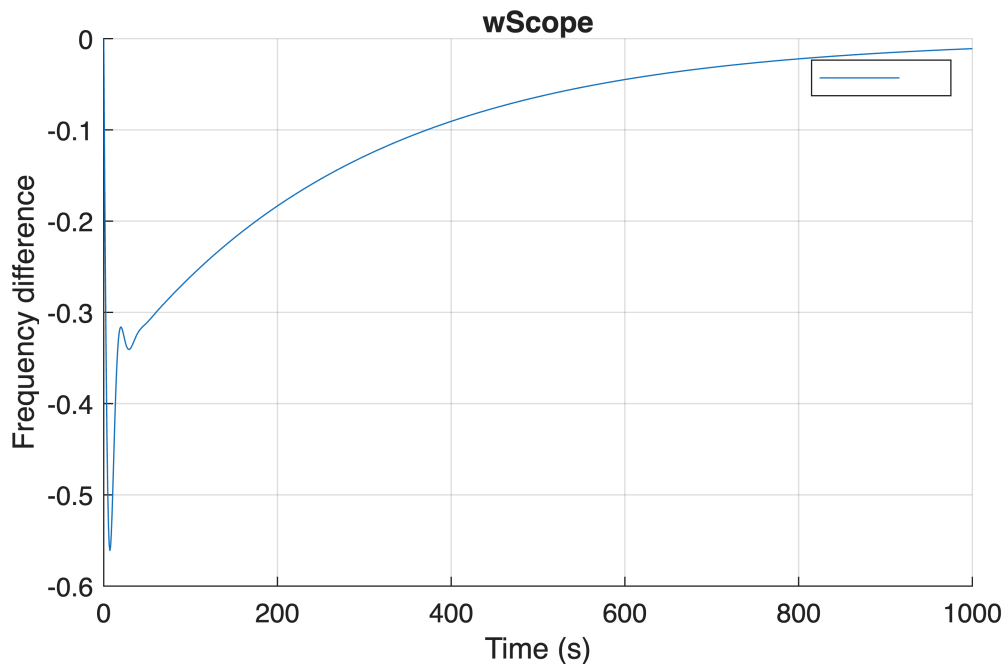
K =
0.2500



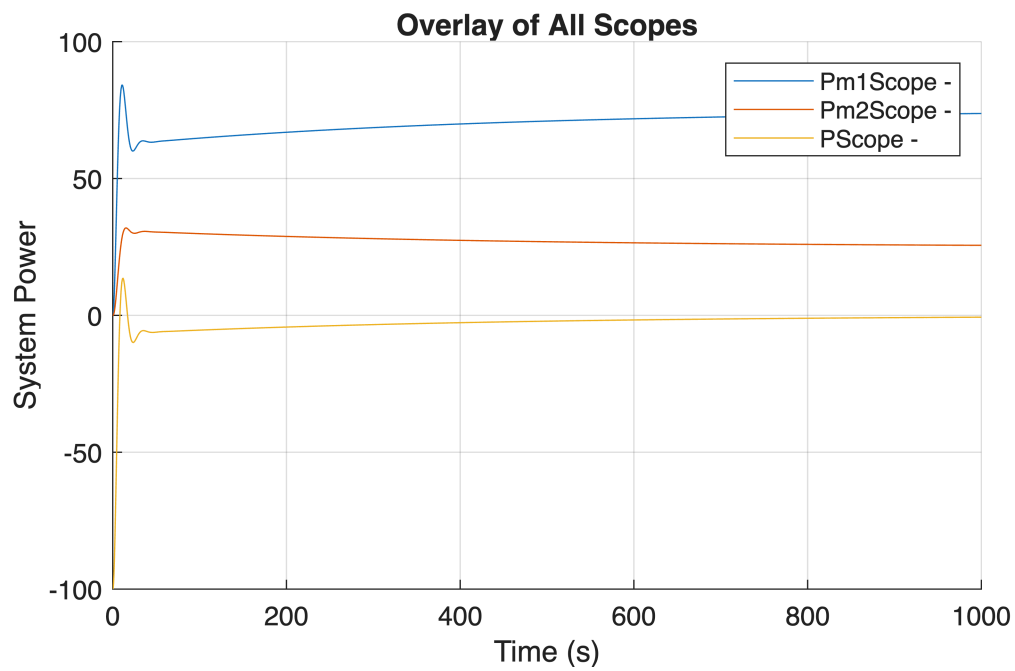
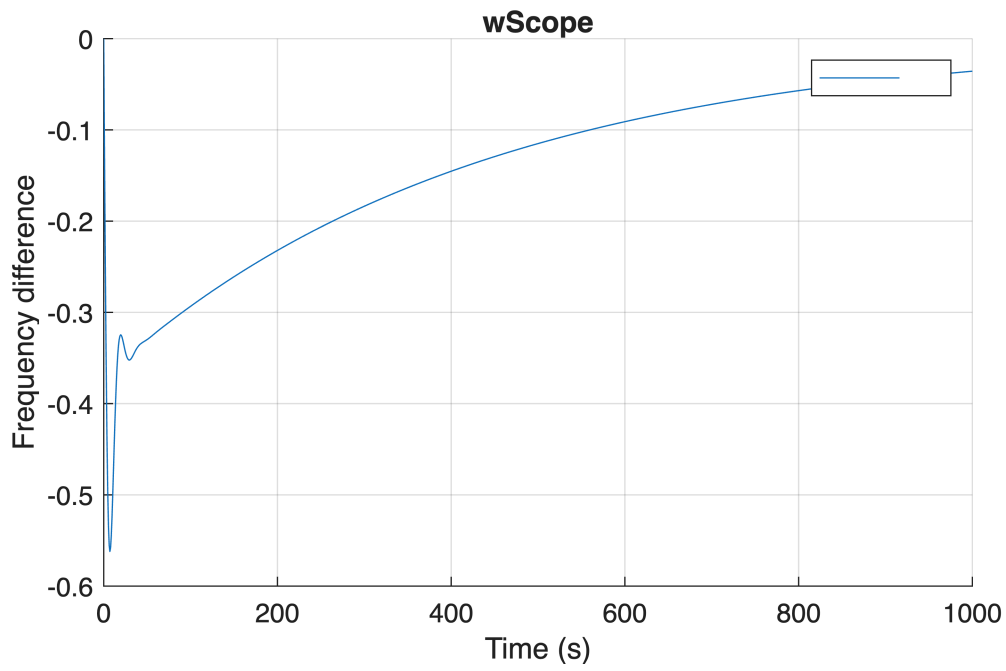
K =
0.2000



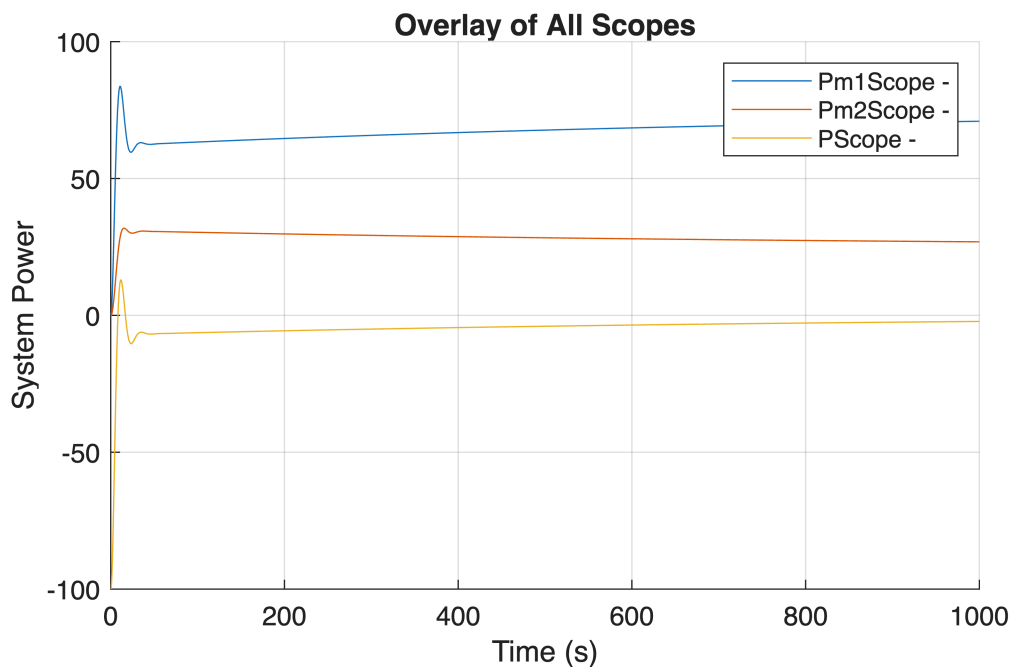
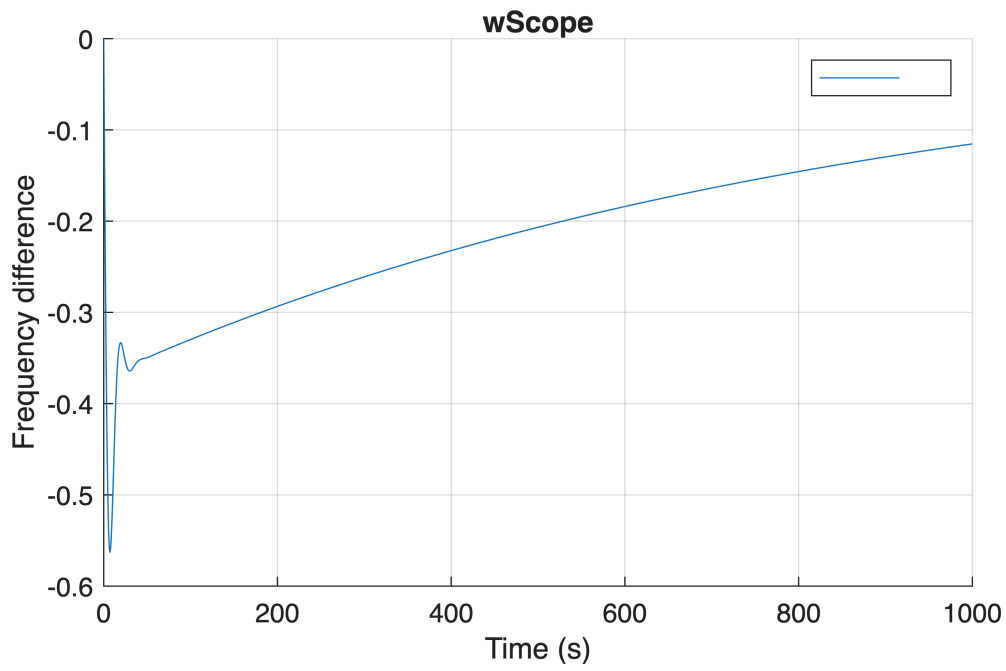
K =
0.1500



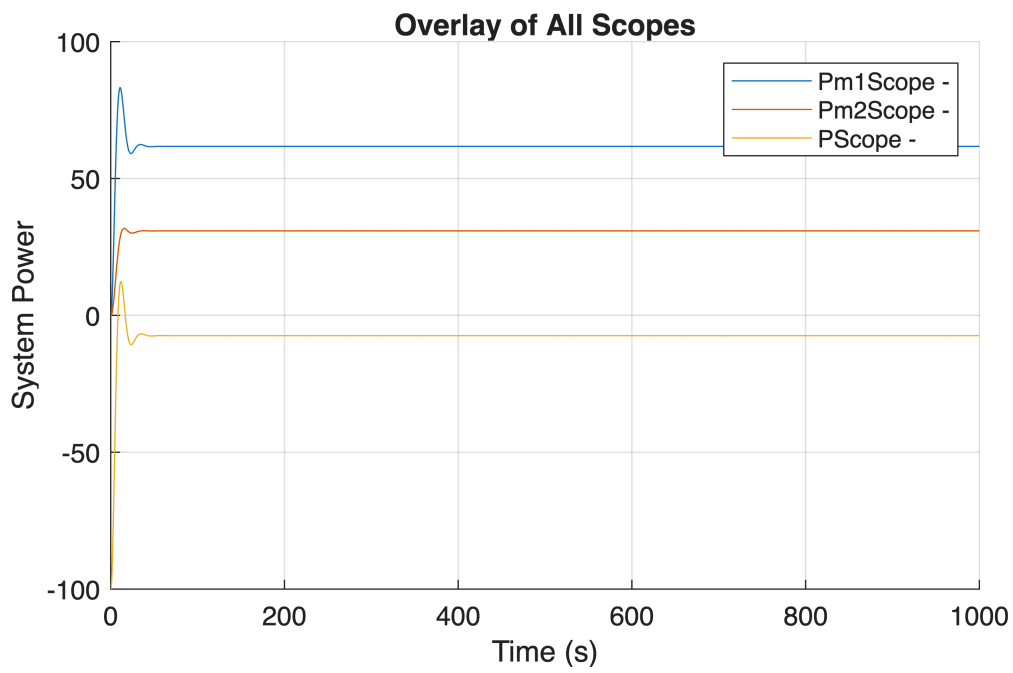
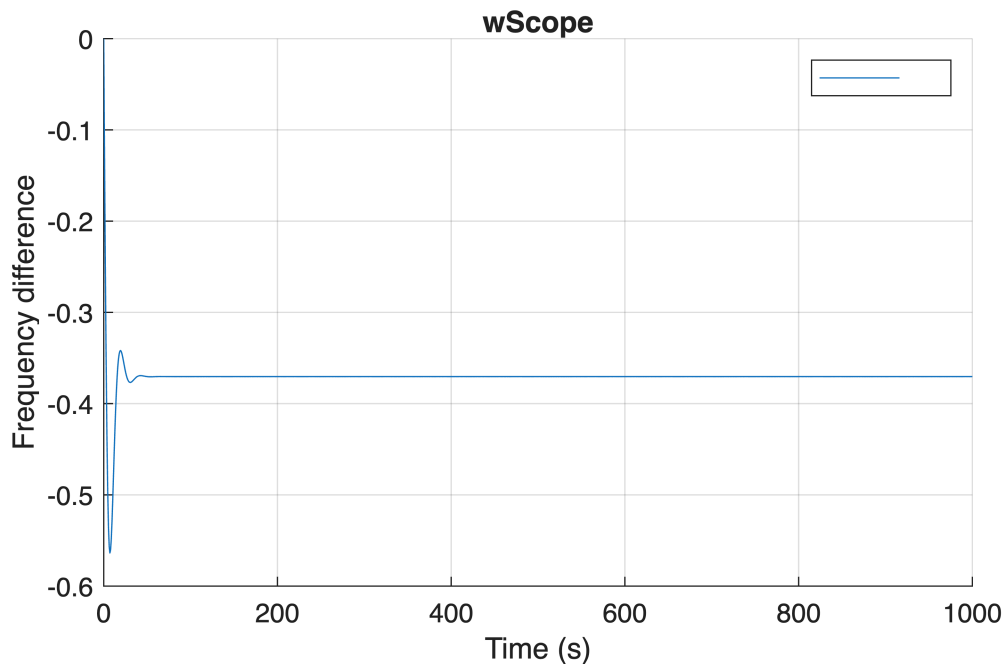
K =
0.1000



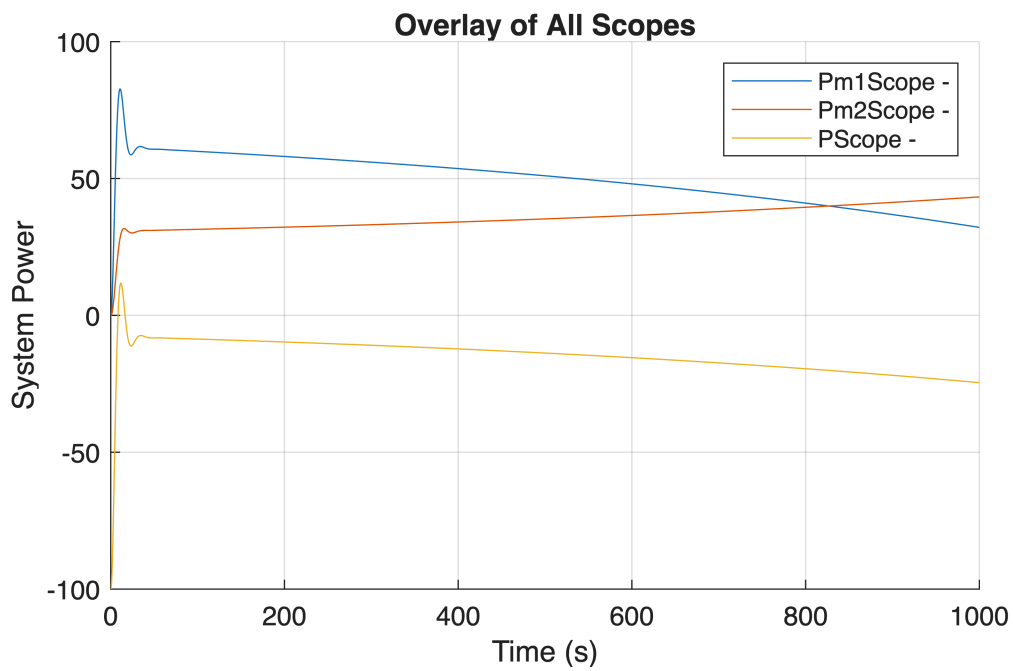
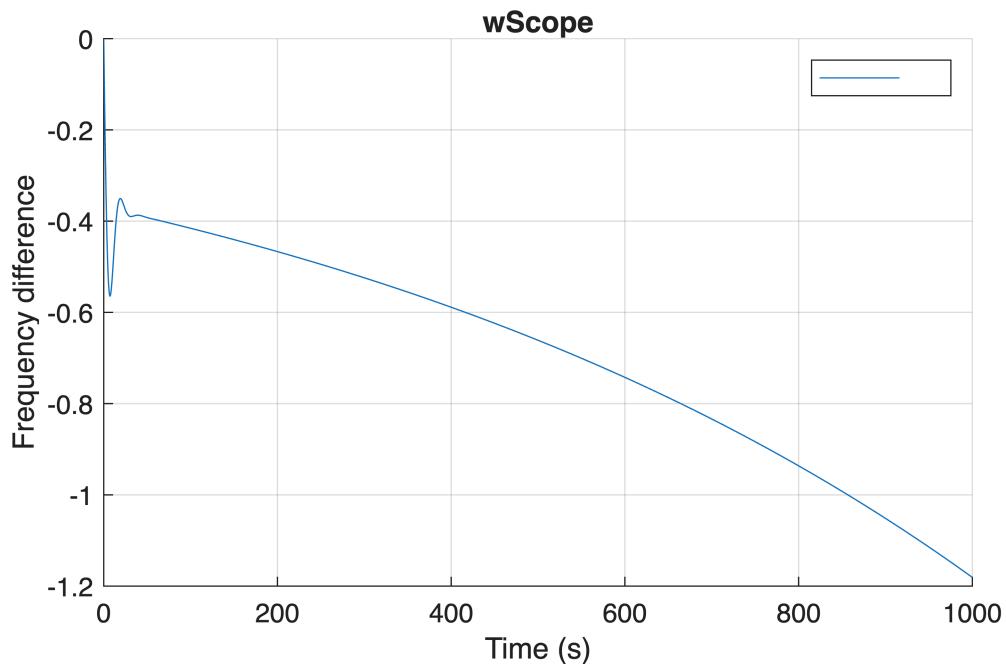
K =
0.0500



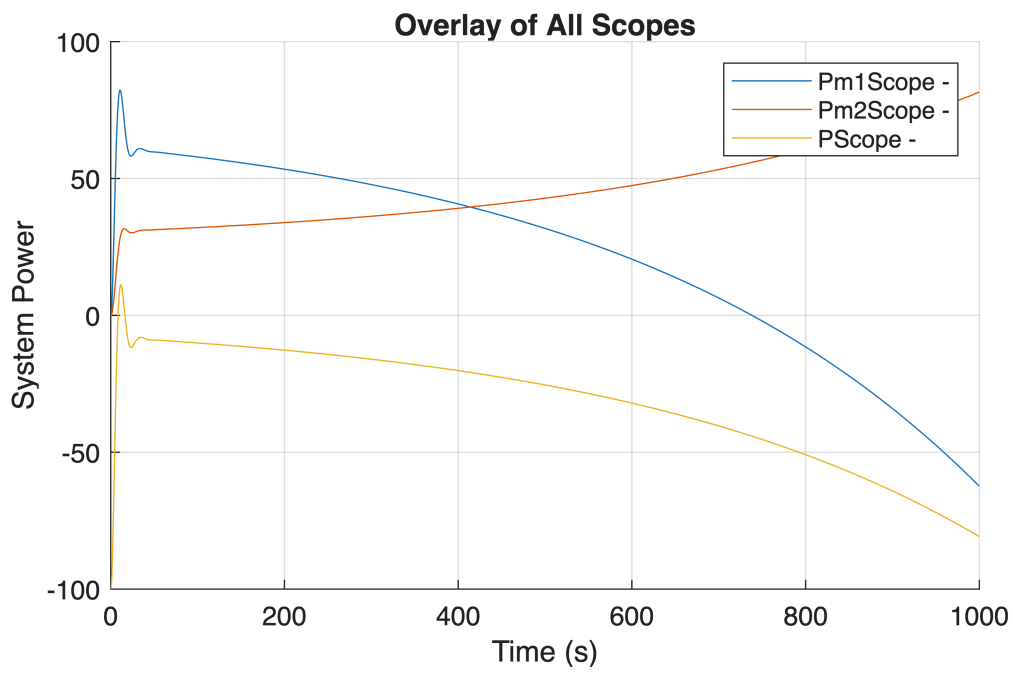
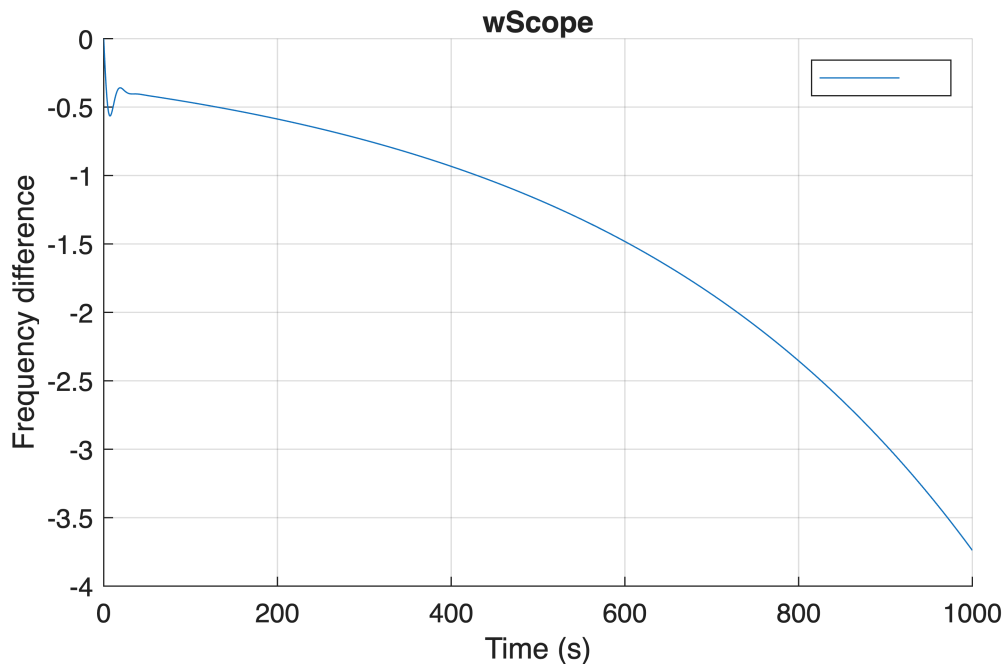
K =
1.3878e-17



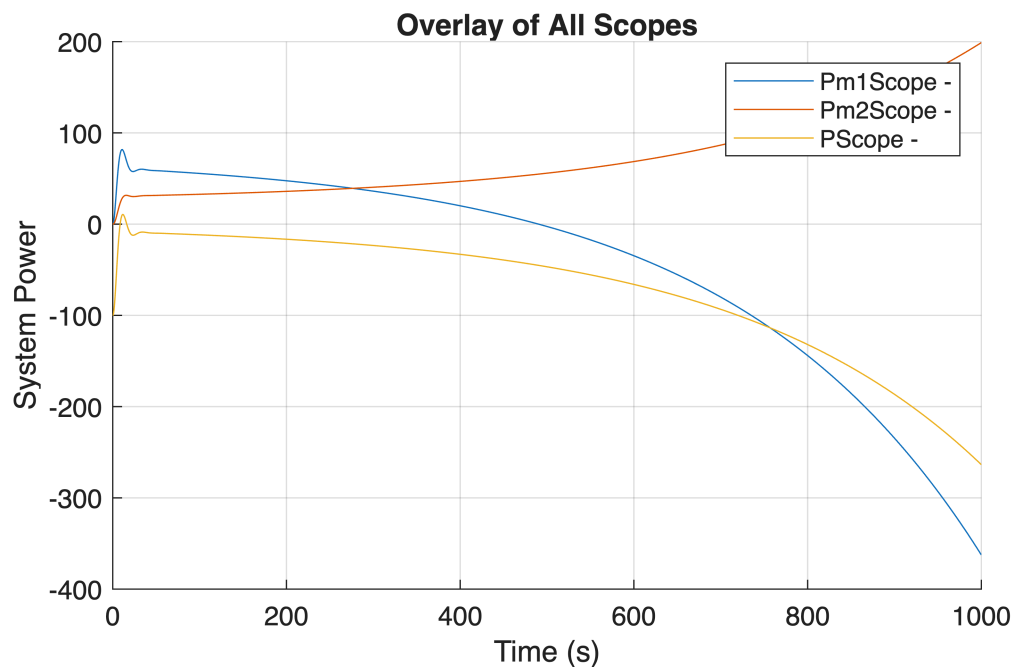
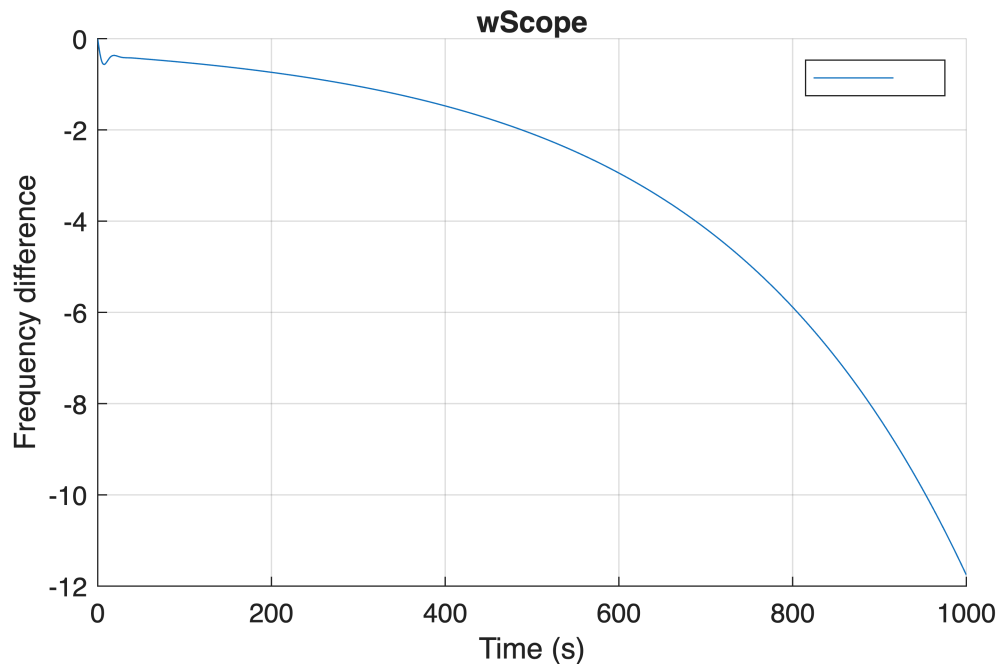
K =
-0.0500



K =
-0.1000



K =
-0.1500



K =
-0.2000

% Question 2
% after selecting K

Tch1 = 3 %s

Tch1 =
3


```
Tch2 = 8 %s
```

```
Tch2 =  
8
```

```
Tg1 = 0.25
```

```
Tg1 =  
0.2500
```

```
Tg2 = 0.50
```

```
Tg2 =  
0.5000
```

```
R1 = 0.0060 * 2 * pi % rad / s MW
```

```
R1 =  
0.0377
```

```
R2 = 0.0120 * 2 * pi % rad / s MW
```

```
R2 =  
0.0754
```

```
M = 1.1141e+08 / 10^6 % convert to MW s^2 / rad
```

```
M =  
111.4100
```

```
D = 20 / ( 2 * pi) % MW s / rad
```

```
D =  
3.1831
```

```
K = .25 %MW
```

```
K =  
0.2500
```

```
% we do the math in MW rads/s
```

```
mdl = 'ass3q2';  
load_system(mdl);  
min15 = 60*15
```

```
min15 =  
900
```

```
set_param(mdl, 'StopTime', '1000');
```

```

%% Run simulation
out = sim mdl);

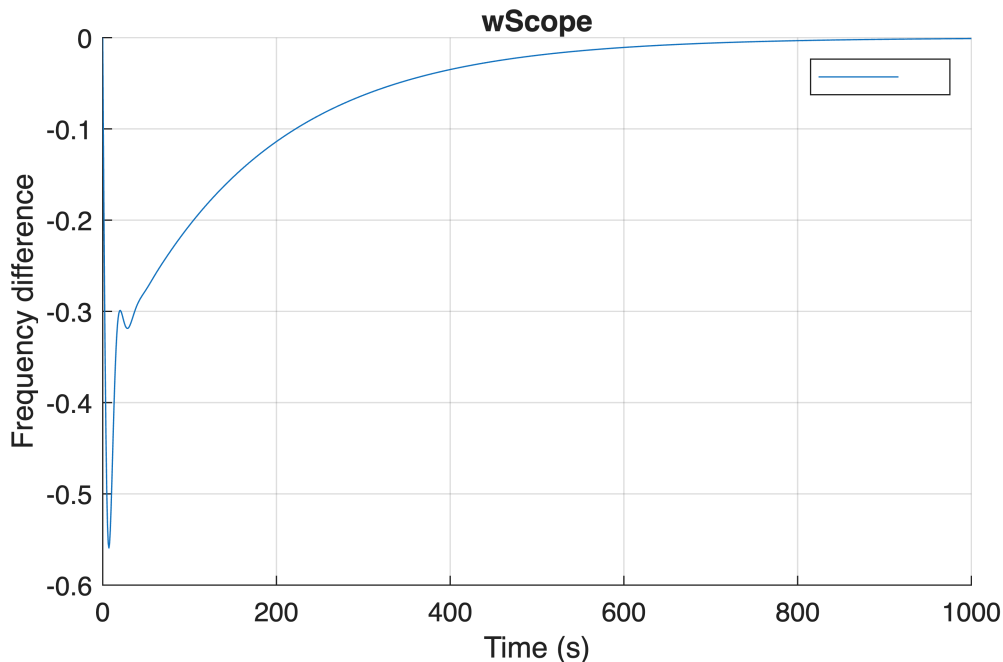
scopeNames = {'wScope'};

for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logsout
        ds = out.logsout.get(name);
    end

    % Plot all signals in this dataset
    figure('Name', name, 'NumberTitle', 'off');
    hold on; grid on;
    for j = 1:ds.numElements
        sig = ds.get(j);
        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', sig.Name);
    end
    xlabel('Time (s)');
    ylabel('Frequency difference');
    title(name);
    legend('show');
end

```



```

scopeNames = {'Pm1Scope', 'Pm2Scope', 'PScope'};

```

```

figure; hold on; grid on;

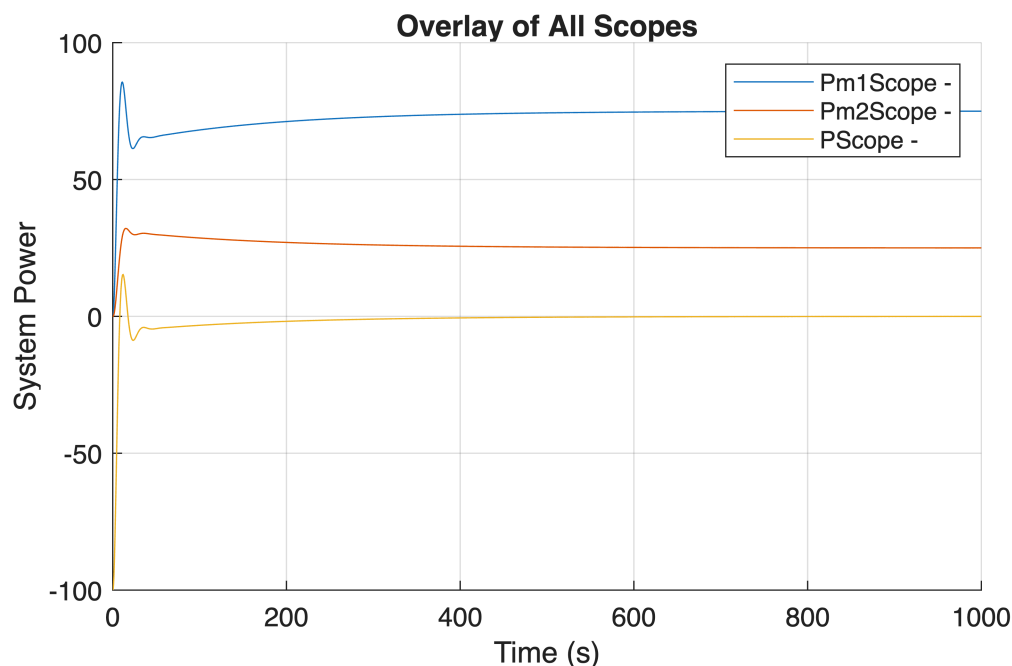
for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logout
        ds = out.logout.get(name);
    end

    % Plot each signal in this scope
    for j = 1:ds.numElements
        sig = ds.get(j);
        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', [name ' - '
sig.Name]);
    end
end

xlabel('Time (s)');
ylabel('System Power');
title('Overlay of All Scopes');
legend('show', 'Interpreter', 'none');

```



```

% Playing around with K, we can stick K = .25 we have one small oscilation
initially
% and then it slowly ramps back to 60 Hz at around 850 seconds

```

```
% there is also minimal ocsilation in the steady state
```

```
% Question 3
```

```
Tch1 = 3 %s
```

```
Tch1 =  
3
```

```
Tch2 = 8 %s
```

```
Tch2 =  
8
```

```
Tg1 = 0.25
```

```
Tg1 =  
0.2500
```

```
Tg2 = 0.50
```

```
Tg2 =  
0.5000
```

```
R1 = 0.0060 * 2 * pi % rad / s MW
```

```
R1 =  
0.0377
```

```
R2 = 0.0120 * 2 * pi % rad / s MW
```

```
R2 =  
0.0754
```

```
Mred = 1.1141e+08 / 10^6 * .6 % convert to MW s^2 / rad
```

```
Mred =  
66.8460
```

```
Dred = 20 / ( 2 * pi) * .5 % MW s / rad
```

```
Dred =  
1.5915
```

```
K = 0.25 %MW
```

```
K =  
0.2500
```

```
% we do the math in MW rads/s
```

```

mdl = 'ass3q3';
load_system(mdl);
set_param(mdl, 'StopTime', '100');

%% Run simulation
out = sim(mdl);

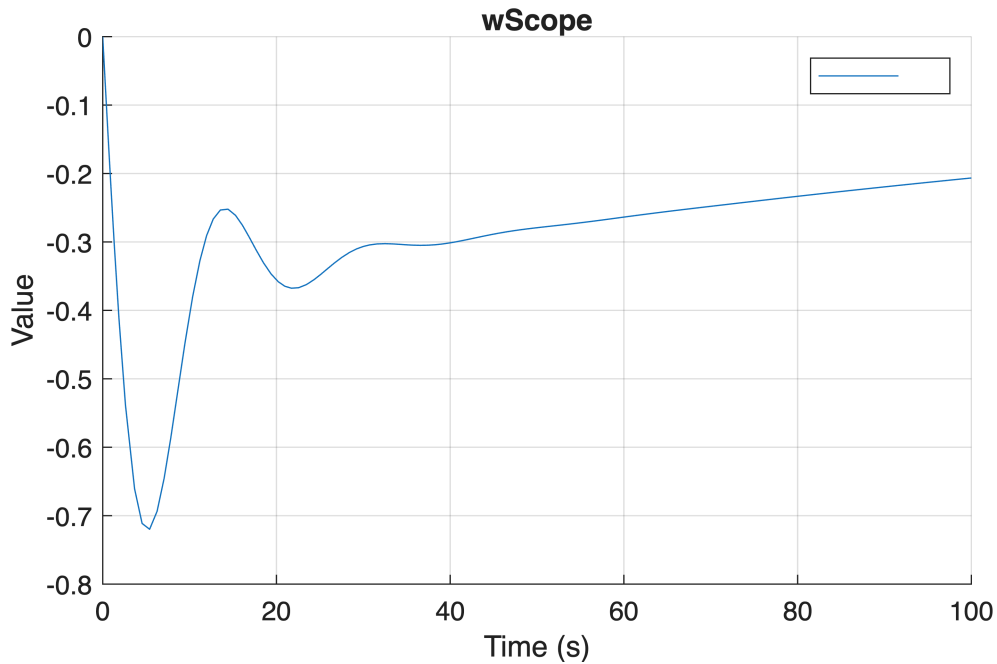
scopeNames = {'wScope'};

for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logouts
        ds = out.logouts.get(name);
    end

    % Plot all signals in this dataset
    figure('Name', name, 'NumberTitle', 'off');
    hold on; grid on;
    for j = 1:ds.numElements
        sig = ds.get(j);
        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', sig.Name);
    end
    xlabel('Time (s)');
    ylabel('Value');
    title(name);
    legend('show');
end
end

```



```
scopeNames = {'Pm1Scope', 'Pm2Scope', 'PScope'};

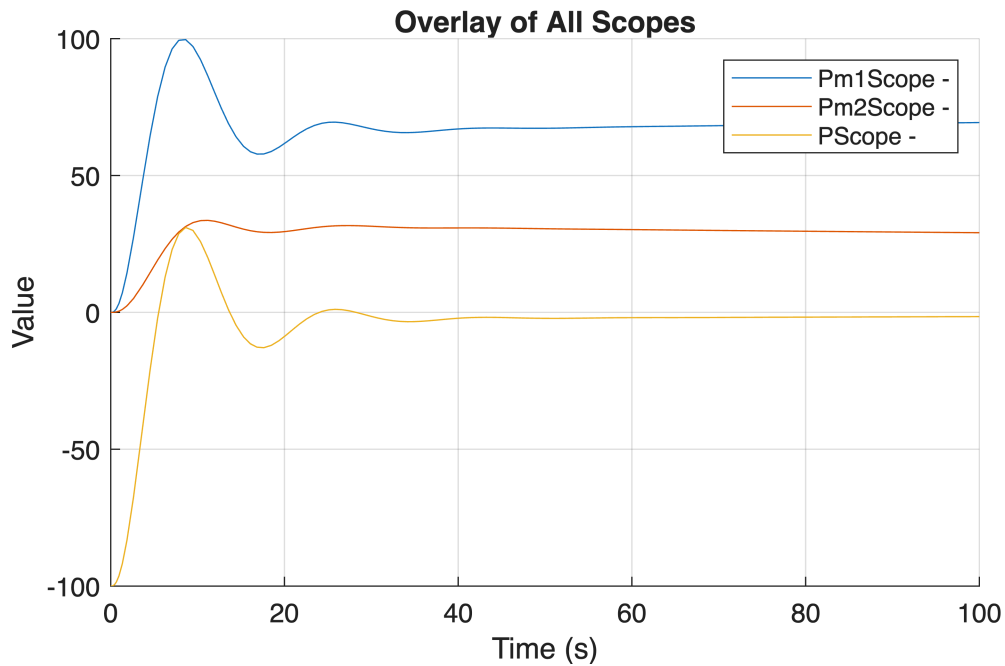
figure; hold on; grid on;

for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logouts
        ds = out.logouts.get(name);
    end

    % Plot each signal in this scope
    for j = 1:ds.numElements
        sig = ds.get(j);
        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', [name ' - '
sig.Name]);
    end
end

xlabel('Time (s)');
ylabel('Value');
title('Overlay of All Scopes');
legend('show', 'Interpreter', 'none');
```



```
% For the frequency, we get a lower Nadir as well as more oscillations.
% the same can be said about power, as Ptotal goes to 25Mw in less than 5
% seconds, then goes to -10 MW and then goes bck above around 25 seconds
% then back below.
% reducing M and D make the drops faster and have more oscillation for the
% same K we selected.
```

```
% Question 4 no control
```

```
Tch1 = 3 %s
```

```
Tch1 =
3
```

```
Tch2 = 8 %s
```

```
Tch2 =
8
```

```
Tg1 = 0.25
```

```
Tg1 =
0.2500
```

```
Tg2 = 0.50
```

```
Tg2 =
0.5000
```

```
R1 = 0.0060 * 2 * pi % rad / s MW
```

```
R1 =  
0.0377
```

```
R2 = 0.0120 * 2 * pi % rad / s MW
```

```
R2 =  
0.0754
```

```
M = 1.1141e+08 / 10^6 % convert to MW s^2 / rad
```

```
M =  
111.4100
```

```
D = 20 / ( 2 * pi) % MW s / rad
```

```
D =  
3.1831
```

```
B = 1/R1 + 1/R2 + D % Mw
```

```
B =  
42.9718
```

```
K = .25
```

```
K =  
0.2500
```

```
T = 20000
```

```
T =  
20000
```

```
% we do the math in MW rads/s
```

```
mdl = 'ass3q4';  
load_system(mdl);  
set_param(mdl, 'StopTime', '1200');
```

```
%% Run simulation  
out = sim(mdl);
```

```
scopeNames = {'wScope', 'wScope1'};
```

```
for i = 1:numel(scopeNames)  
    name = scopeNames{i};
```

```
    % Try to get dataset directly from 'out'  
    try  
        ds = out.get(name);
```

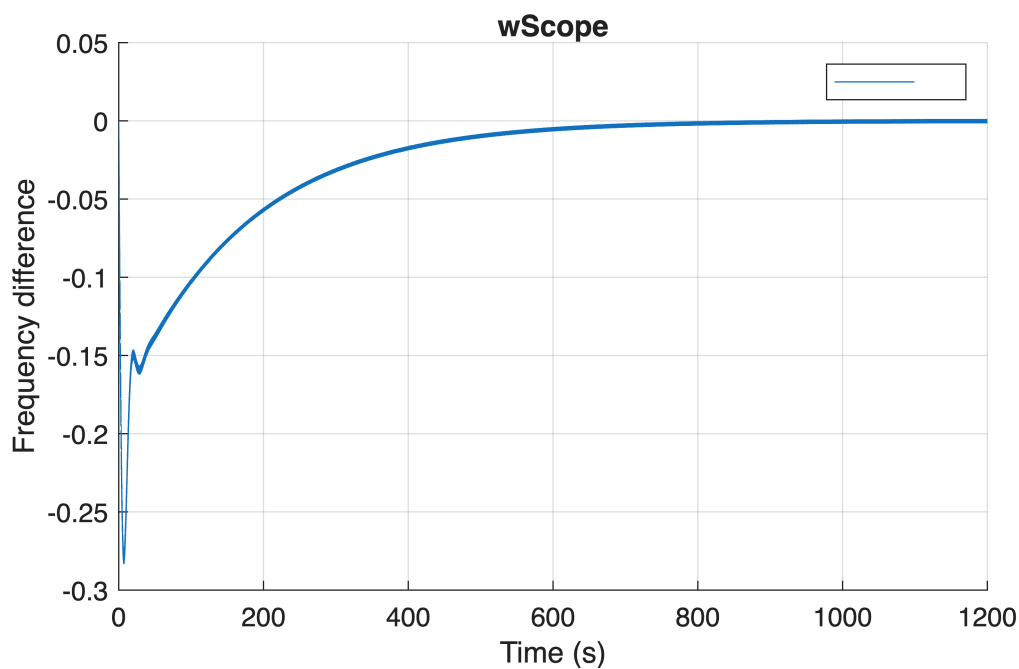


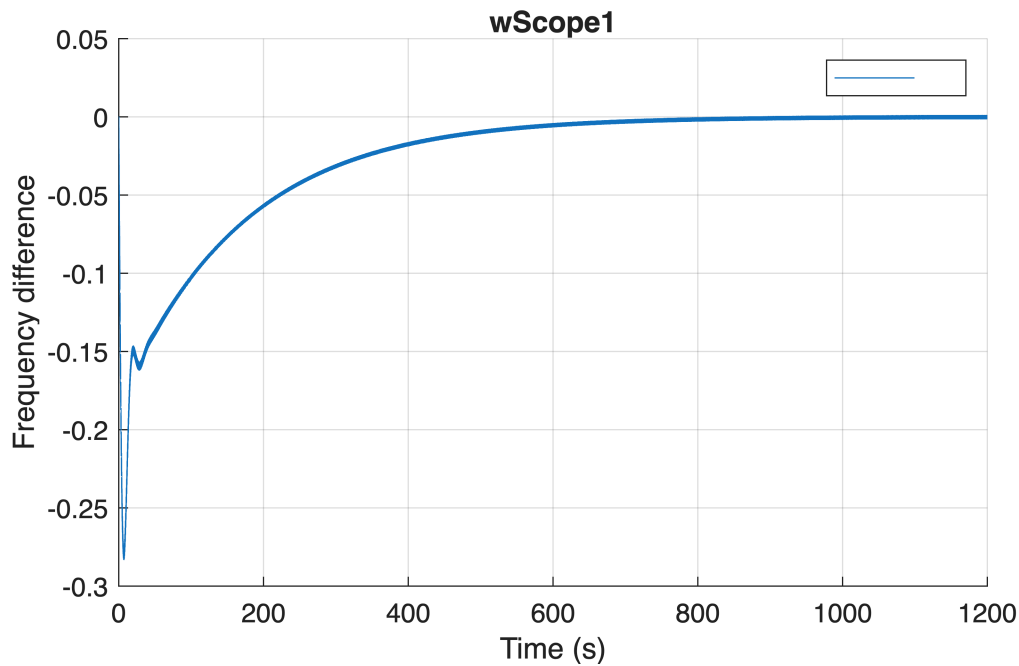
```

catch
    % Otherwise get it from logout
    ds = out.logout.get(name);
end

% Plot all signals in this dataset
figure('Name', name, 'NumberTitle', 'off');
hold on; grid on;
for j = 1:ds.numElements
    sig = ds.get(j);
    plot(sig.Values.Time, sig.Values.Data, 'DisplayName', sig.Name);
end
xlabel('Time (s)');
ylabel('Frequency difference');
title(name);
legend('show');
end

```





```
scopeNames = {'Pm1Scope', 'Pm2Scope', 'PScope'};

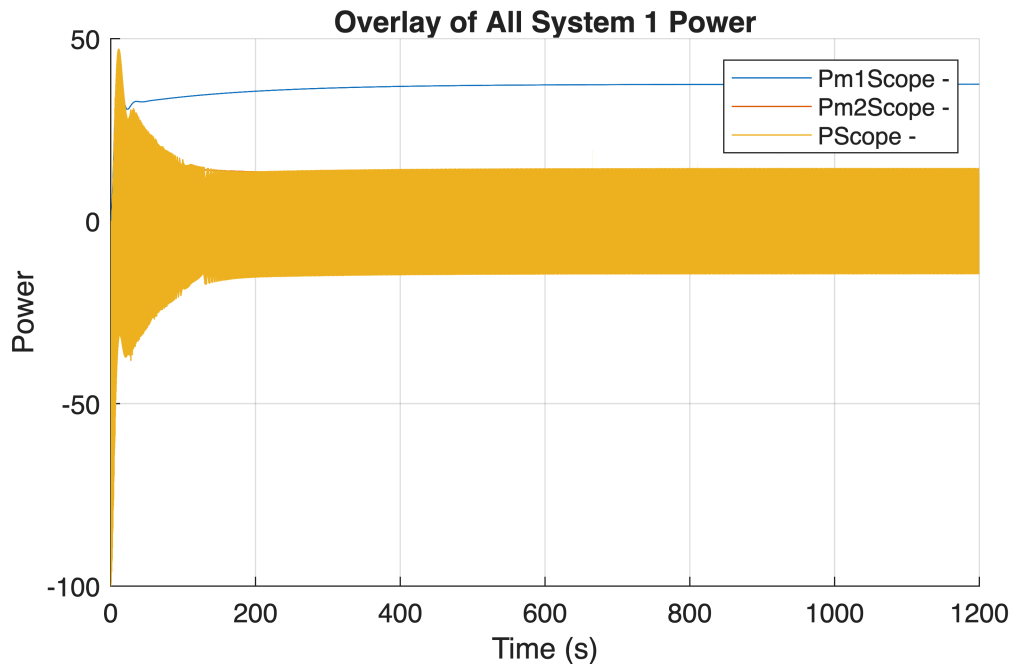
figure; hold on; grid on;

for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logouts
        ds = out.logouts.get(name);
    end

    % Plot each signal in this scope
    for j = 1:ds.numElements
        sig = ds.get(j);
        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', [name ' - '
sig.Name]);
    end
end

xlabel('Time (s)');
ylabel('Power');
title('Overlay of All System 1 Power');
legend('show', 'Interpreter', 'none');
```



```
scopeNames = {'Pm1Scope1','Pm2Scope1','PScope1'};

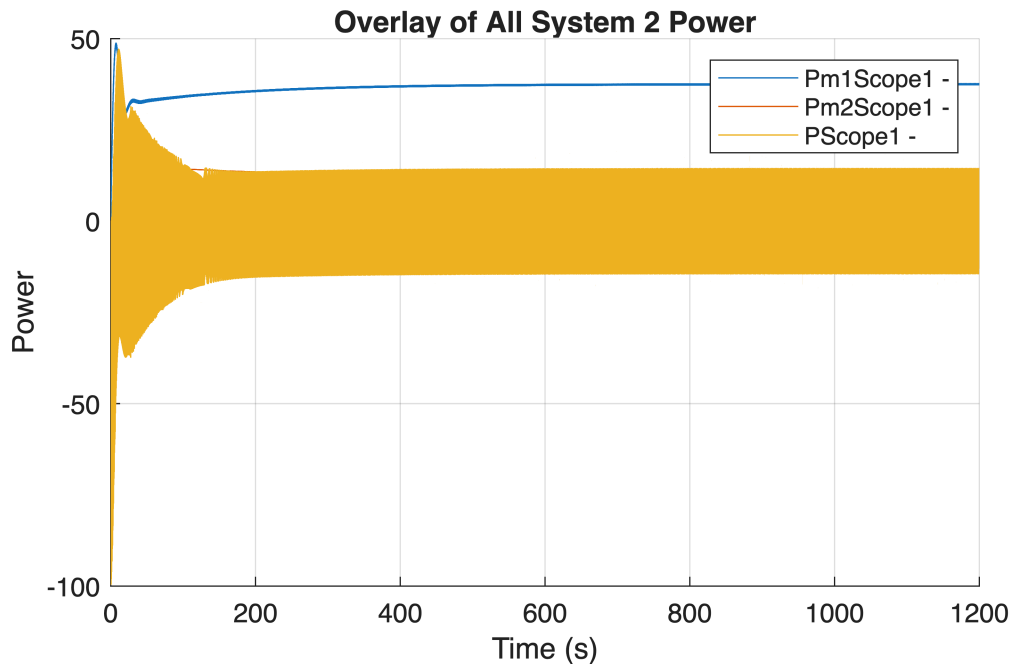
figure; hold on; grid on;

for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logouts
        ds = out.logouts.get(name);
    end

    % Plot each signal in this scope
    for j = 1:ds.numElements
        sig = ds.get(j);
        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', [name ' - '
sig.Name]);
    end
end

xlabel('Time (s)');
ylabel('Power');
title('Overlay of All System 2 Power');
legend('show', 'Interpreter', 'none');
```



```
scopeNames = {'PTieLine'};

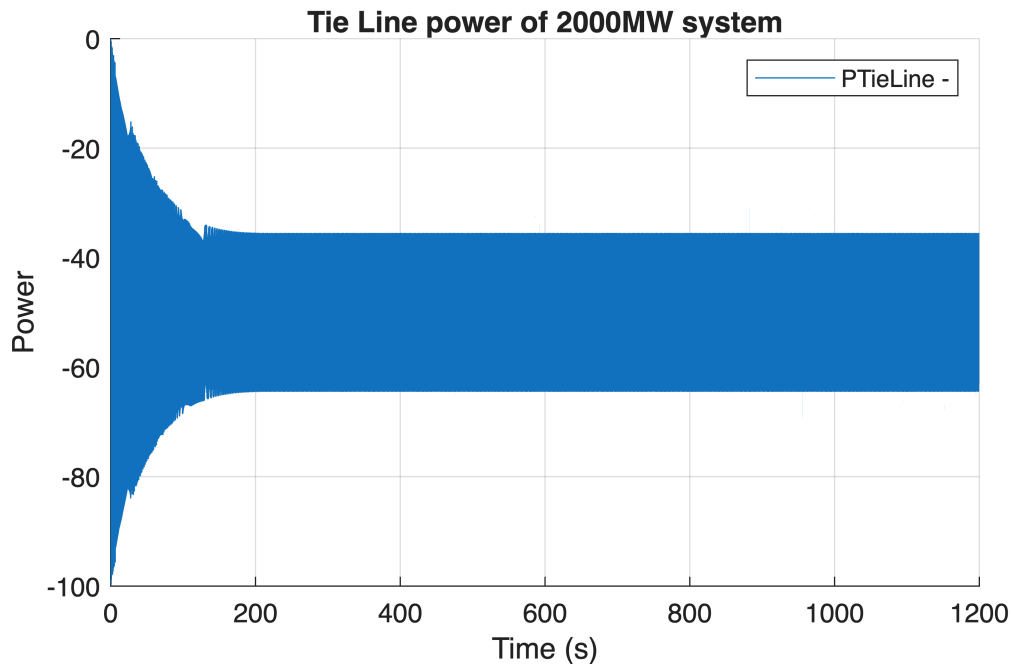
figure; hold on; grid on;

for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logout
        ds = out.logout.get(name);
    end

    % Plot each signal in this scope
    for j = 1:ds.numElements
        sig = ds.get(j);
        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', [name ' - '
sig.Name]);
    end
end

xlabel('Time (s)');
ylabel('Power');
title('Tie Line power of 2000MW system');
legend('show', 'Interpreter', 'none');
```



% We can see that the frequency goes back to 0, but the tie line power
% oscillates around -50 MW.

% Question 4 with control

Tch1 = 3 %s

Tch1 =
3

Tch2 = 8 %s

Tch2 =
8

Tg1 = 0.25

Tg1 =
0.2500

Tg2 = 0.50

Tg2 =
0.5000

R1 = 0.0060 * 2 * pi % rad / s MW

R1 =
0.0377

R2 = 0.0120 * 2 * pi % rad / s MW

```
R2 =  
0.0754
```

```
M = 1.1141e+08 / 10^6 % convert to MW s^2 / rad
```

```
M =  
111.4100
```

```
D = 20 / ( 2 * pi) % MW s / rad
```

```
D =  
3.1831
```

```
B = 1/R1 + 1/R2 + D % Mw
```

```
B =  
42.9718
```

```
K = .1
```

```
K =  
0.1000
```

```
T = 20000
```

```
T =  
20000
```

```
% we do the math in MW rads/s
```

```
mdl = 'ass3q4Control';  
load_system(mdl);  
set_param(mdl, 'StopTime', '1200');
```

```
%% Run simulation  
out = sim(mdl);
```

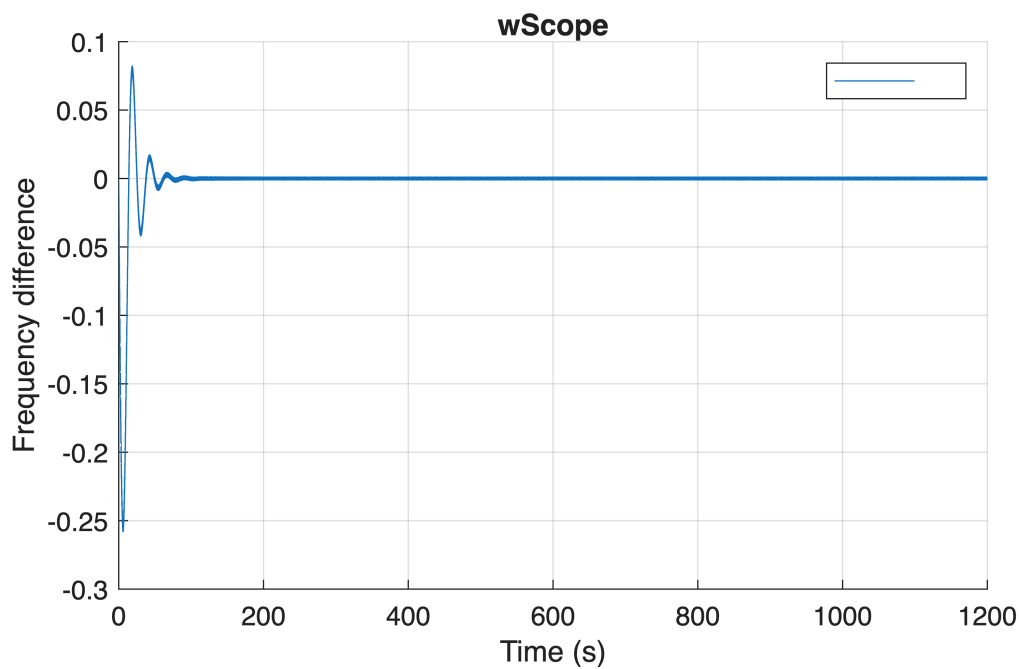
```
scopeNames = {'wScope', 'wScope1'};
```

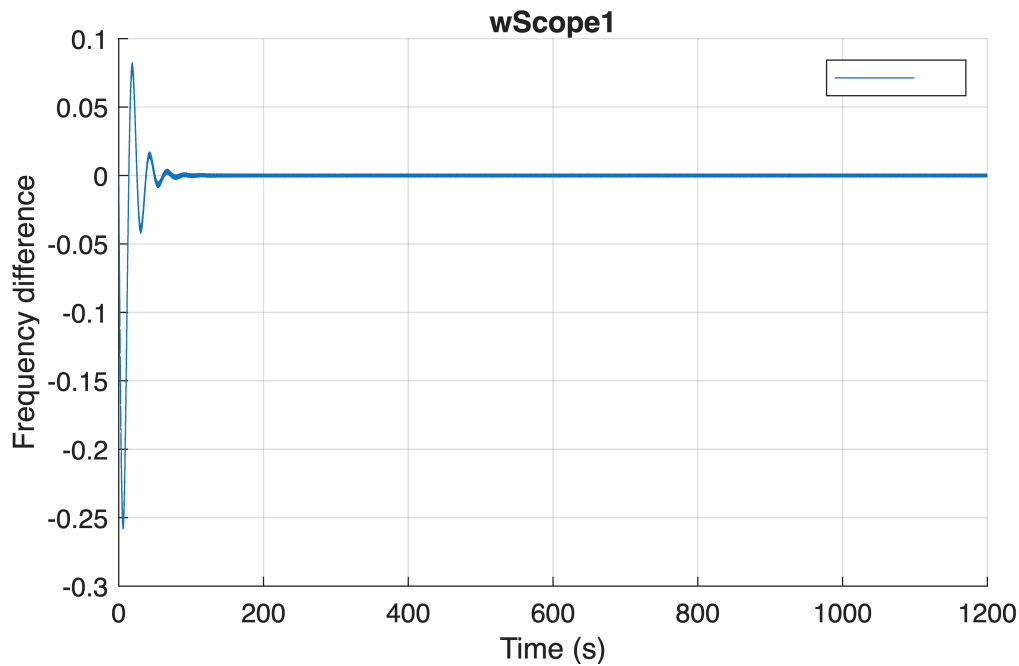
```
for i = 1:numel(scopeNames)  
    name = scopeNames{i};  
  
    % Try to get dataset directly from 'out'  
    try  
        ds = out.get(name);  
    catch  
        % Otherwise get it from logout  
        ds = out.logout.get(name);  
    end
```

```

% Plot all signals in this dataset
figure('Name', name, 'NumberTitle', 'off');
hold on; grid on;
for j = 1:ds.numElements
    sig = ds.get(j);
    plot(sig.Values.Time, sig.Values.Data, 'DisplayName', sig.Name);
end
xlabel('Time (s)');
ylabel('Frequency difference');
title(name);
legend('show');
end

```





```
scopeNames = {'Pm1Scope', 'Pm2Scope', 'PScope'};

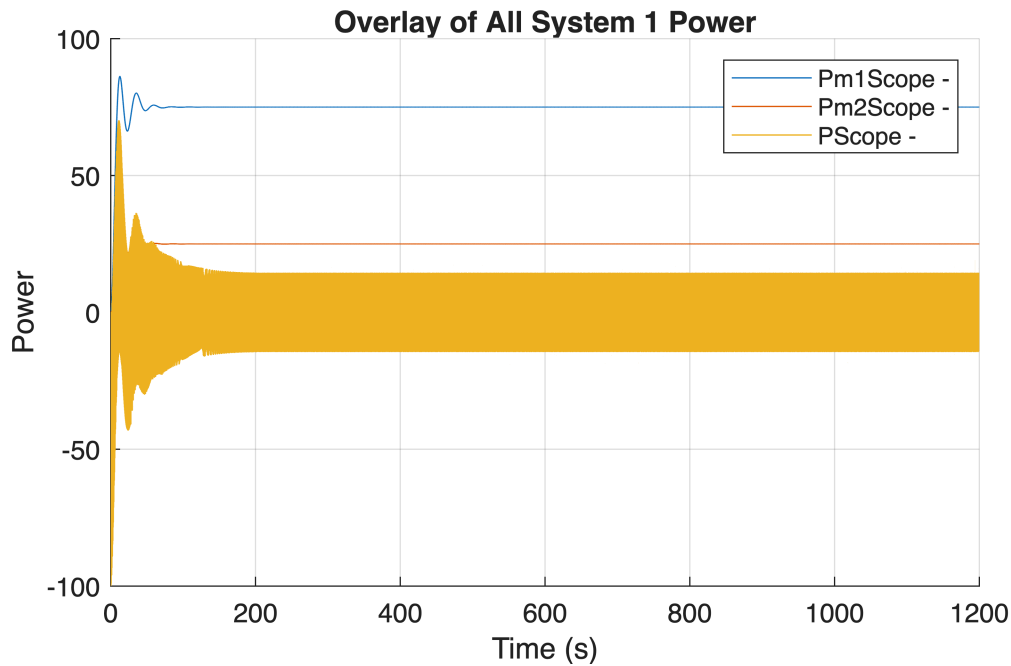
figure; hold on; grid on;

for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logouts
        ds = out.logouts.get(name);
    end

    % Plot each signal in this scope
    for j = 1:ds.numElements
        sig = ds.get(j);
        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', [name ' - '
sig.Name]);
    end
end

xlabel('Time (s)');
ylabel('Power');
title('Overlay of All System 1 Power');
legend('show', 'Interpreter', 'none');
```

```
scopeNames = {'Pm1Scope1', 'Pm2Scope1', 'PScope1'};

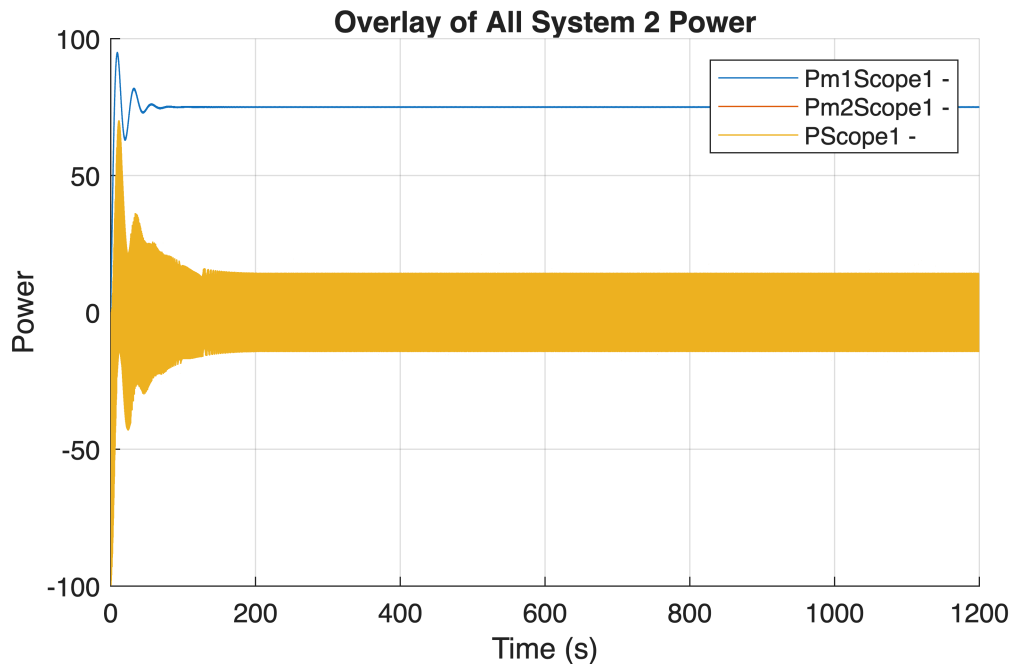
figure; hold on; grid on;

for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logouts
        ds = out.logouts.get(name);
    end

    % Plot each signal in this scope
    for j = 1:ds.numElements
        sig = ds.get(j);
        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', [name ' - '
sig.Name]);
    end
end

xlabel('Time (s)');
ylabel('Power');
title('Overlay of All System 2 Power');
legend('show', 'Interpreter', 'none');
```



```
scopeNames = {'PTieLine'};

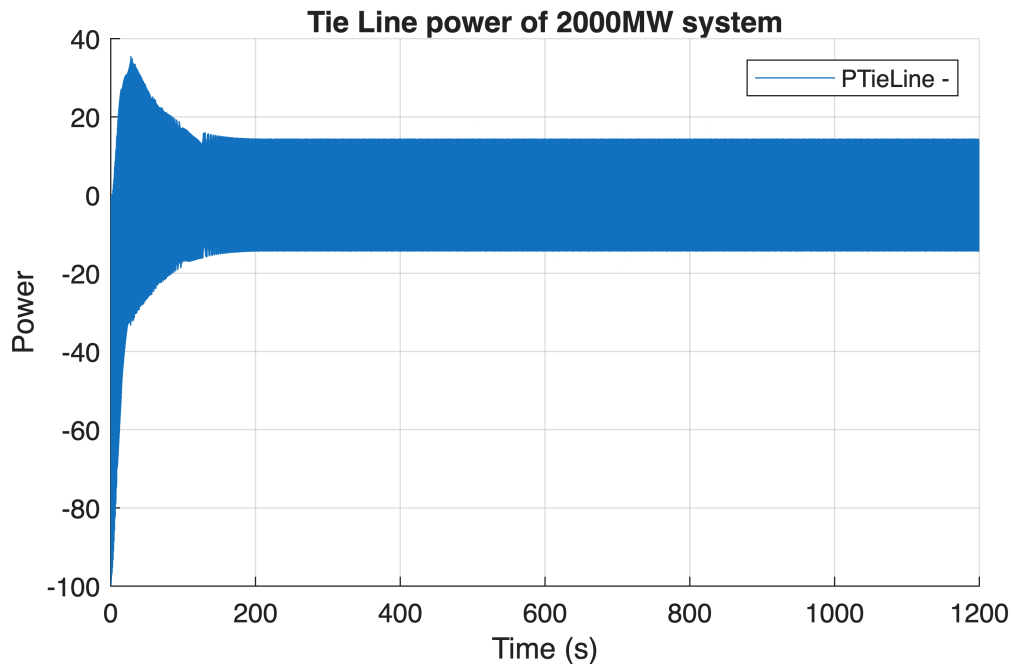
figure; hold on; grid on;

for i = 1:numel(scopeNames)
    name = scopeNames{i};

    % Try to get dataset directly from 'out'
    try
        ds = out.get(name);
    catch
        % Otherwise get it from logouts
        ds = out.logouts.get(name);
    end

    % Plot each signal in this scope
    for j = 1:ds.numElements
        sig = ds.get(j);
        plot(sig.Values.Time, sig.Values.Data, 'DisplayName', [name ' - '
sig.Name]);
    end
end

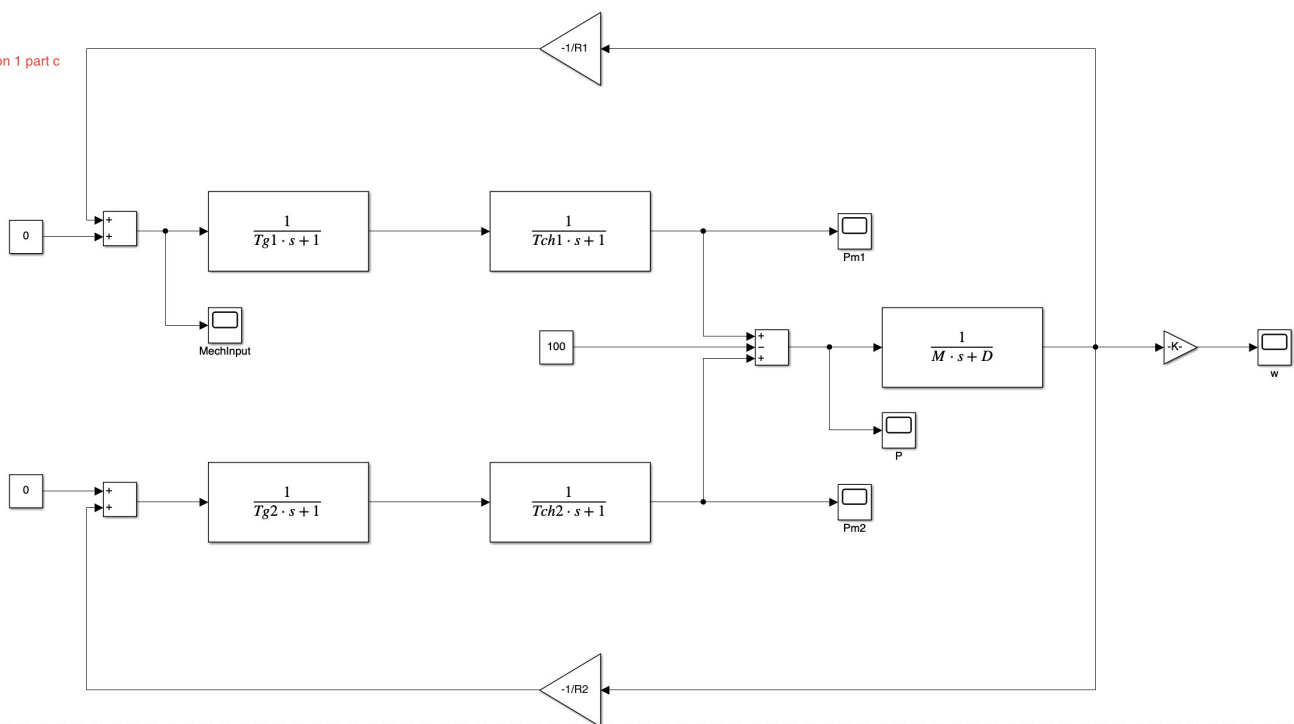
xlabel('Time (s)');
ylabel('Power');
title('Tie Line power of 2000MW system');
legend('show', 'Interpreter', 'none');
```



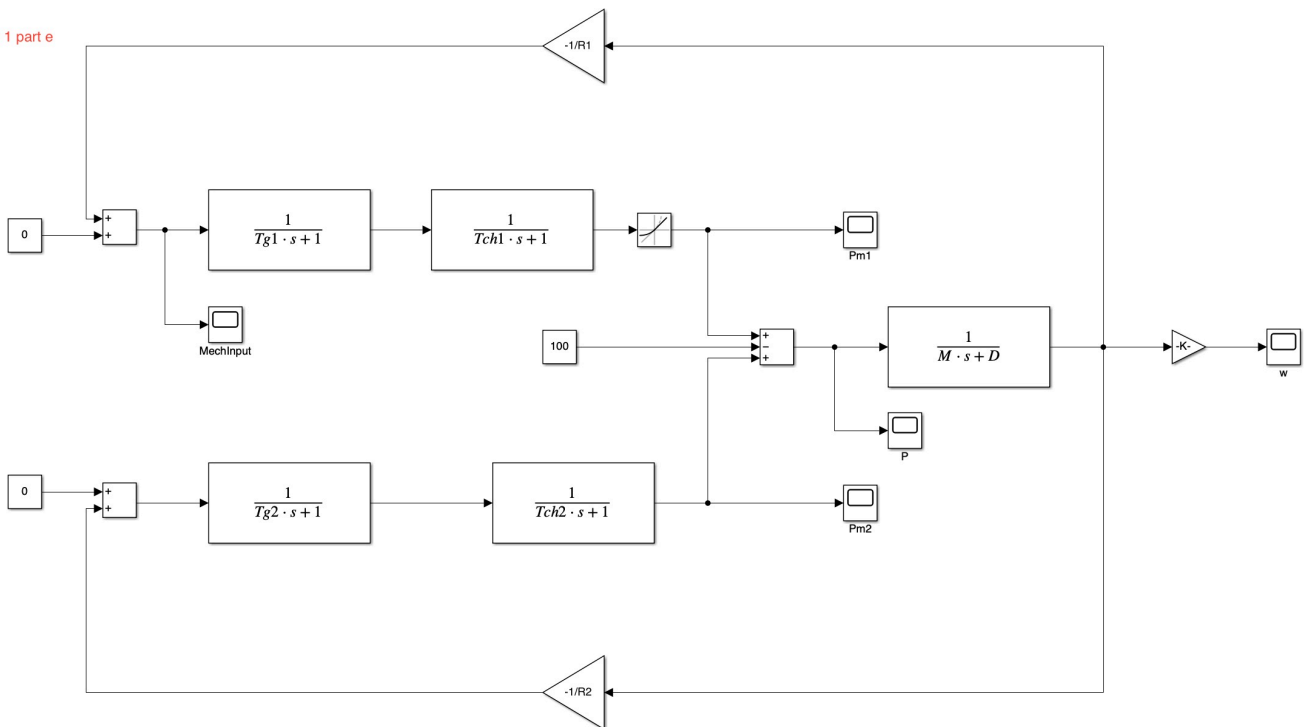
% Adding additional controls, we are able to get the frequency to 0 and have
 % the tie line Power stabilize and oscillate around 0
 % some problems we notice is how quick the responses are and the
 % oscillations.
 % These can be mitigated by adding some delays and ramp controls, as well
 % as fine tuning the control paramters of the model further.

```
% R1 = 0.0060 * 2 * pi % rad / s MW
% R2 = 0.0120 * 2 * pi % rad / s MW
%
%
% M = 1.1141e+08 / 10^6 % convert to MW s^2 / rad
% D = 20 / ( 2 * pi) % MW s / rad
%
% B = 1/R1 + 1/R2 + D
%
% T = 20000
% K = 0.01
```

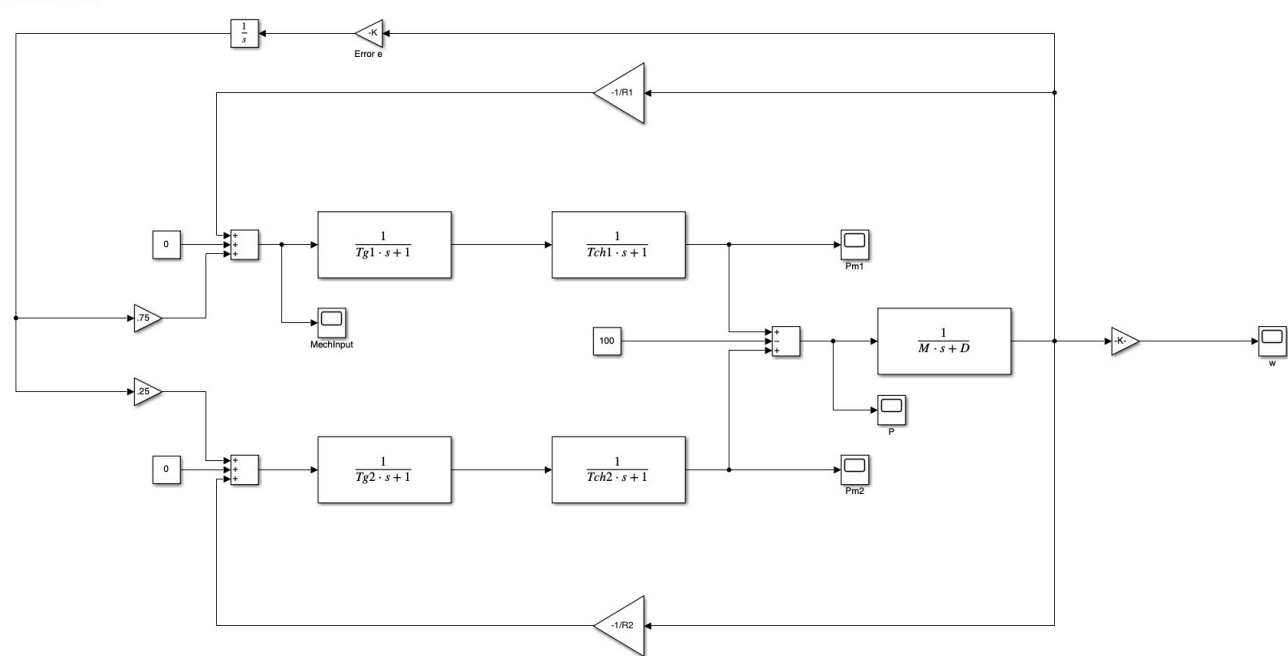
Question 1 part c



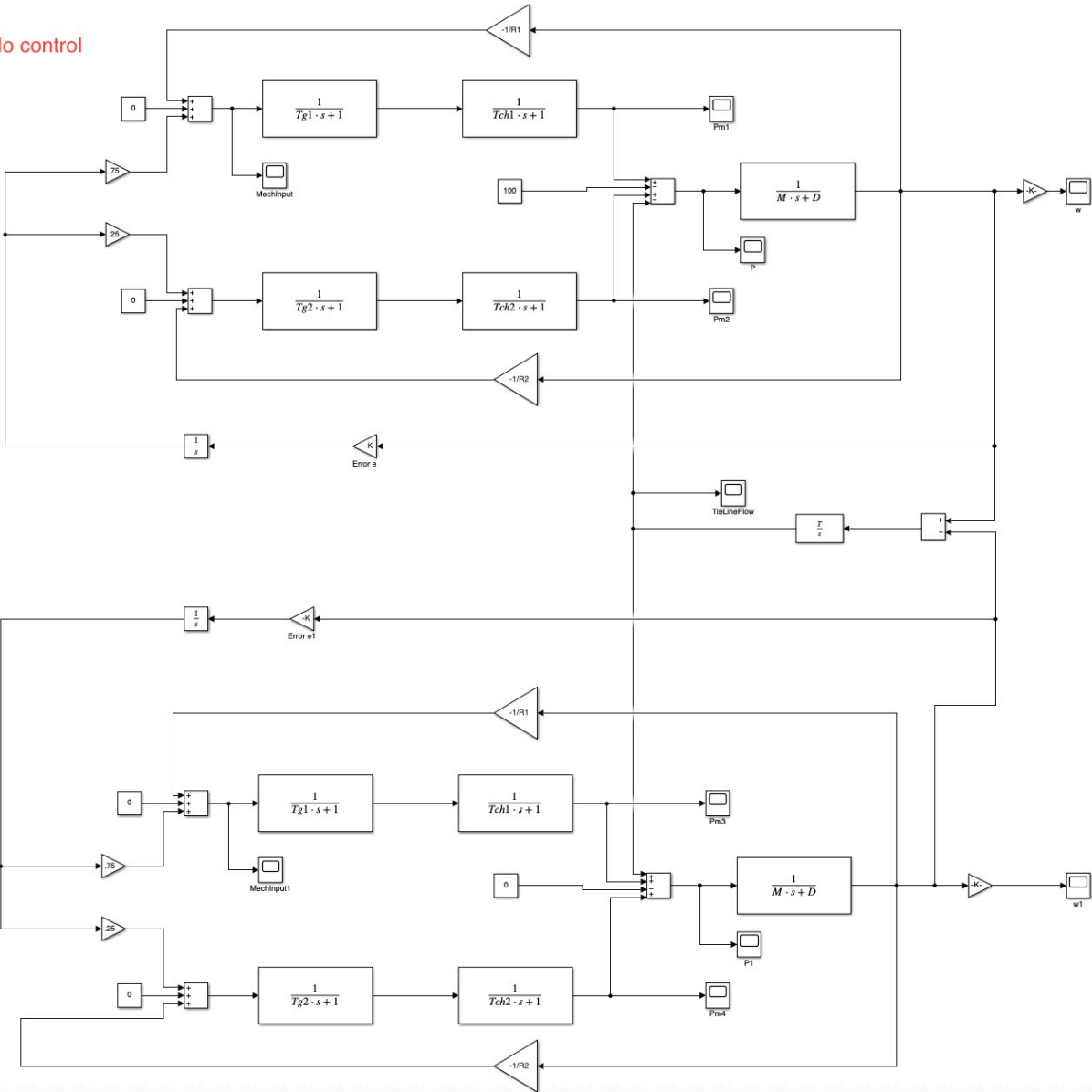
Question 1 part e



Question 2



Question 4 No control



Question 4 with control

